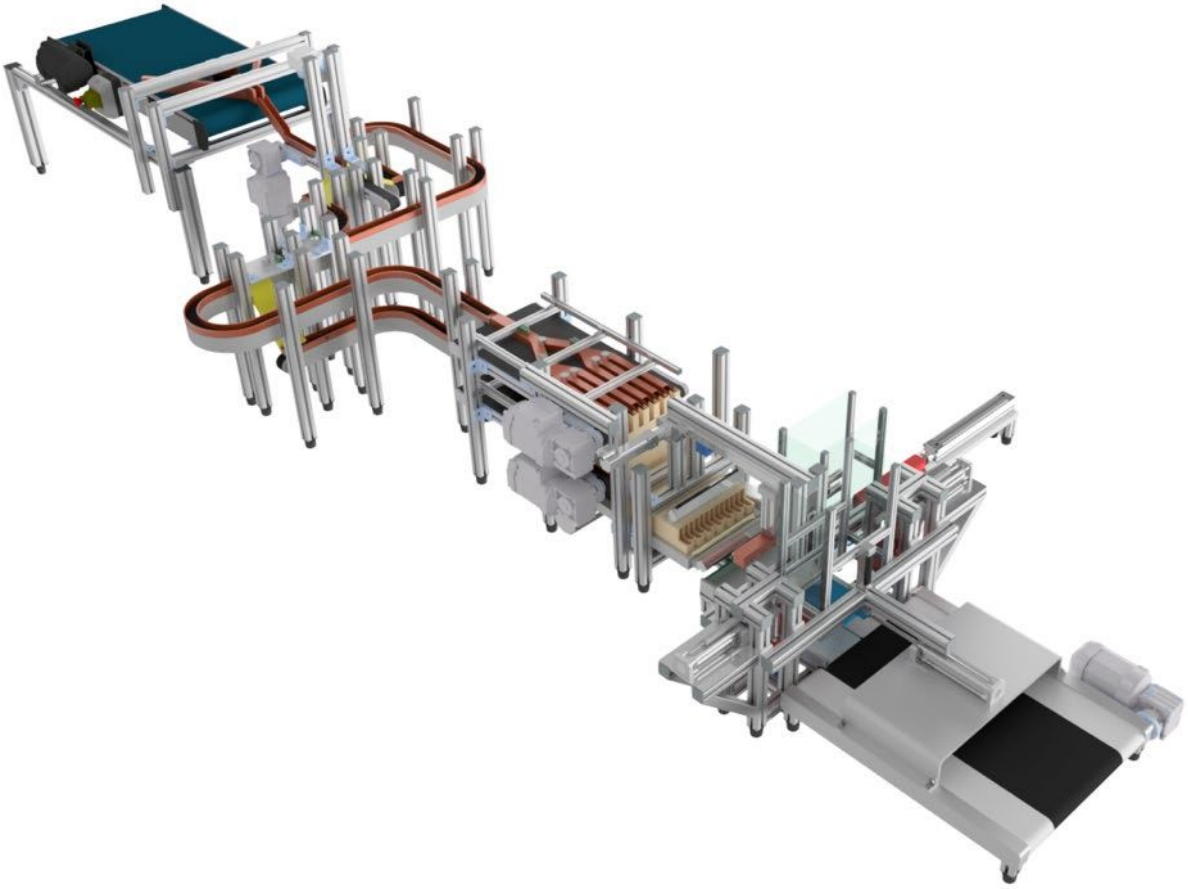
ME20025 – Design 2023/2024

Machine Design Documentation Report

Machine Design Control Rod Module Assembly for Nuclear Reactors

Group 25



Luka Cherriman (lc2557), Luca Piergentili (lp2022), William Chappell (wc645), Leon Roman (lr811)

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Introduction

The aim of this project was to design a machine concept capable of assembling control rod modules for use in a nuclear reactor. Per the client's request, control rods were to be inputted into the machine through a 500 mm wide conveyor in a disorganised manner. It was the objective of the machine to organise and pack the control rods in the correct arrangement into two carriers to form a module. To meet the annual quota of the client, the assembly of a module should take no more than 180 seconds. The machine should run for 24 hours a day for up to 10 years. Our final concept achieved these requirements in the following steps: first, the control rods would be funnelled to form an organised feed. The machine then scans and separates the rods based on their orientation (plug or socket end). At this point the machine splits into two levels. Both levels consist of splitters. A total of 13 splitters would be used to organise the rods in their respective positions ready to be assembled into the carriers. Once the bottom carriers are inputted onto the assembly platform, the rods are inserted. The upper carrier is then pressed onto the half-assembled module. The finished product is subsequently sent out on a 500 mm wide belt. The entire module assembly process takes 136.4 seconds, and the machine has a total cost of £20,746 (excluding labour costs).

Concept Evaluation and Evolution

We started off the design process by brainstorming several functions our machine should have. Using a Morphological Chart (Figure 1), each member came up with a design for each function. Some ideas could complete multiple functions simultaneously, and we found that some functions could be done in parallel with others. Each team member came up with a concept sketch to help visualise our individual propositions (Figures 3 to 6). This helped us analyse and pick out the most realistic and effective concepts. We later decided which ones were the most adapted to the design objectives, basing ourselves on the needs and wants expressed in our product design specification (Figure 49). The MCDA (Figure 2) helped us decide which design ideas we would put together to come up with our final machine concept. Instead of evaluating the concepts as wholes, we evaluated each individual function by itself as the final concept could be an amalgamation of ideas from the various concepts. Some of the ideas were further developed to make it more functional and adapted to our final machine design. The ideas we chose were those that would allow our machine to achieve the objectives we had set it in addition to those that were set in the hand-out. We wanted our machine to separate rods based on their orientation, facilitate their insertion into the carriers, and determine a way to insert multiple rods simultaneously. We believe that our final machine concept is the most apt to do this, using each of our design ideas.

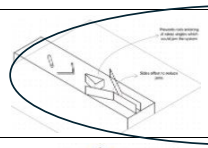
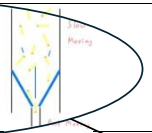
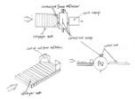
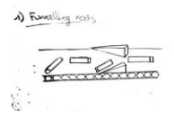
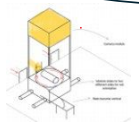
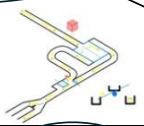
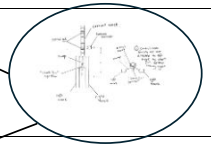
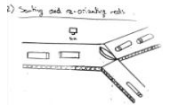
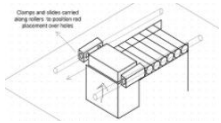
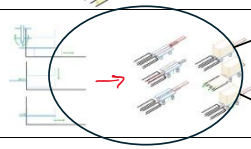
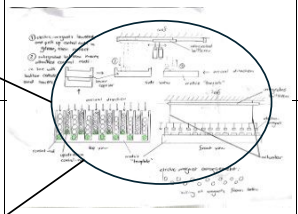
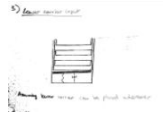
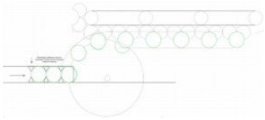
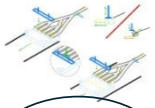
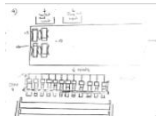
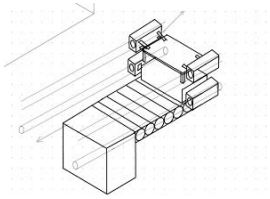
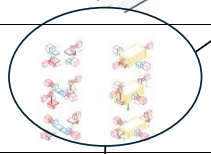
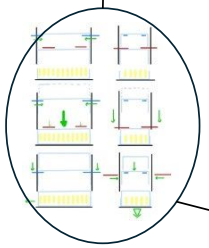
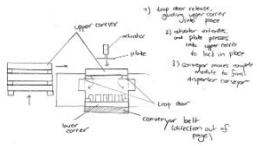
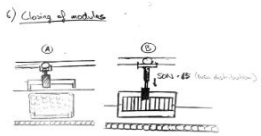
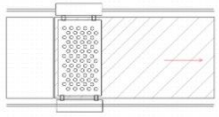
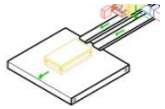
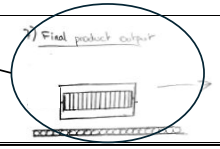
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1) Funnelling Rods				
2) Sorting (scanning) and re-orienting (flipping) rods				
3) Lower carrier retreval				
4) Lower carrier rod placement				
5) Upper carrier retreval				
6) Aligning upper carrier with rods				
7) Upper carrier secured to rods				
8) Final product output				

Figure 1. *Morphological Chart*

		1				2				3				4			
1) Funnelling Rods	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	0	0		4	1	4		4	0	0	
	B	2	0	0		2	0	0		2	-1	-2		2	0	0	
	C	1	0	0		1	0	0		1	0	0		1	0	0	
	D	3	0	0		3	0	0		3	0	0		3	0	0	
	E	5	0	0		5	0	0		5	-1	-5		5	-1	-5	
	F	3	0	0		3	0	0		3	0	0		3	0	0	
	G	4	0	0		4	0	0		4	0	0		4	-1	-4	
	H	3	0	0		3	0	0		3	-1	-3		3	0	0	
		Total		0		Total		0		Total		-6		Total		-9	
2) Sorting (scanning) and re-orienting (flipping) rods	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	1	4		4	1	4		4	1	4	
	B	2	0	0		2	1	2		2	1	2		2	1	2	
	C	1	0	0		1	0	0		1	0	0		1	0	0	
	D	3	0	0		3	-1	-3		3	1	3		3	1	3	
	E	5	0	0		5	1	5		5	1	5		5	-1	-5	
	F	3	0	0		3	0	0		3	0	0		3	-1	-3	
	G	4	0	0		4	0	0		4	0	0		4	0	0	
	H	3	0	0		3	0	0		3	0	0		3	1	3	
		Total		0		Total		8		Total		14		Total		4	
3) Lower carrier retreval	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	1	4		4		0		4	-1	-4	
	B	2	0	0		2	0	0		2		0		2	1	2	
	C	1	0	0		1	0	0		1		0		1	1	1	
	D	3	0	0		3	-1	-3		3		0		3	0	0	
	E	5	0	0		5	1	5		5		0		5	-1	-5	
	F	3	0	0		3	0	0		3		0		3	0	0	
	G	4	0	0		4	0	0		4		0		4	1	4	
	H	3	0	0		3	0	0		3		0		3	0	0	
		Total		0		Total		6		Total		0		Total		-2	
4) Lower carrier rod placement	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	0	0		4	1	4		4	0	0	
	B	2	0	0		2	0	0		2	0	0		2	-1	-2	
	C	1	0	0		1	0	0		1	1	1		1	0	0	
	D	3	0	0		3	-1	-3		3	1	3		3	0	0	
	E	5	0	0		5	1	5		5	1	5		5	-1	-5	
	F	3	0	0		3	1	3		3	1	3		3	0	0	
	G	4	0	0		4	0	0		4	1	4		4	0	0	
	H	3	0	0		3	-1	-3		3	0	0		3	-1	-3	
		Total		0		Total		2		Total		20		Total		-10	
5) Upper carrier retreval	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4		0		4	1	4		4	-1	-4		4	-1	-4	
	B	2		0		2	0	0		2	0	0		2	-1	-2	
	C	1		0		1	1	1		1	0	0		1	0	0	
	D	3		0		3	1	3		3	1	3		3	1	3	
	E	5		0		5	1	5		5	1	5		5	-1	-5	
	F	3		0		3	0	0		3	0	0		3	0	0	
	G	4		0		4	1	4		4	1	4		4	0	0	
	H	3		0		3	1	3		3	1	3		3	-1	-3	
		Total		0		Total		20		Total		11		Total		-11	
6) Aligning upper carrier with rods	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	1	4		4	1	4		4	1	4	
	B	2	0	0		2	0	0		2	0	0		2	-1	-2	
	C	1	0	0		1	1	1		1	0	0		1	0	0	
	D	3	0	0		3	1	3		3	0	0		3	0	0	
	E	5	0	0		5	1	5		5	1	5		5	1	5	
	F	3	0	0		3	0	0		3	0	0		3	0	0	
	G	4	0	0		4	1	4		4	0	0		4	0	0	
	H	3	0	0		3	1	3		3	1	3		3	1	3	
		Total		0		Total		20		Total		12		Total		10	
7) Upper carrier secured to rods	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	0	0		4	-1	-4		4	1	4	
	B	2	0	0		2	1	2		2	1	2		2	1	2	
	C	1	0	0		1	1	1		1	0	0		1	0	0	
	D	3	0	0		3	1	3		3	0	0		3	1	3	
	E	5	0	0		5	1	5		5	-1	-5		5	-1	-5	
	F	3	0	0		3	0	0		3	0	0		3	0	0	
	G	4	0	0		4	1	4		4	1	4		4	1	4	
	H	3	0	0		3	1	3		3	0	0		3	-1	-3	
		Total		0		Total		18		Total		-3		Total		5	
8) Final product output	Requirement	Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score		Weight	Score	Weighted Score	
	A	4	0	0		4	1	4		4		0		4	1	4	
	B	2	0	0		2	1	2		2		0		2	1	2	
	C	1	0	0		1	1	1		1		0		1	1	1	
	D	3	0	0		3	1	3		3		0		3	1	3	
	E	5	0	0		5	1	5		5		0		5	1	5	
	F	3	0	0		3	1	3		3		0		3	1	3	
	G	4	0	0		4	1	4		4		0		4	1	4	
	H	3	0	0		3	1	3		3		0		3	1	3	
		Total		0		Total		25		Total		0		Total		25	

Figure 2A. *MCDA*

#	Requirements	Ranking	Weight
A	Speed of Operation	2	4
B	Use of Standard Parts	7	2
C	Energy Consumption	8	1
D	Footprint / Size	5	3
E	Reliability	1	5
F	Rod Safety/ care	4	3
G	Maintenence and Operator Safety	3	4
H	Cost	6	3

Figure 23B. *MCDA Requirement weighting*

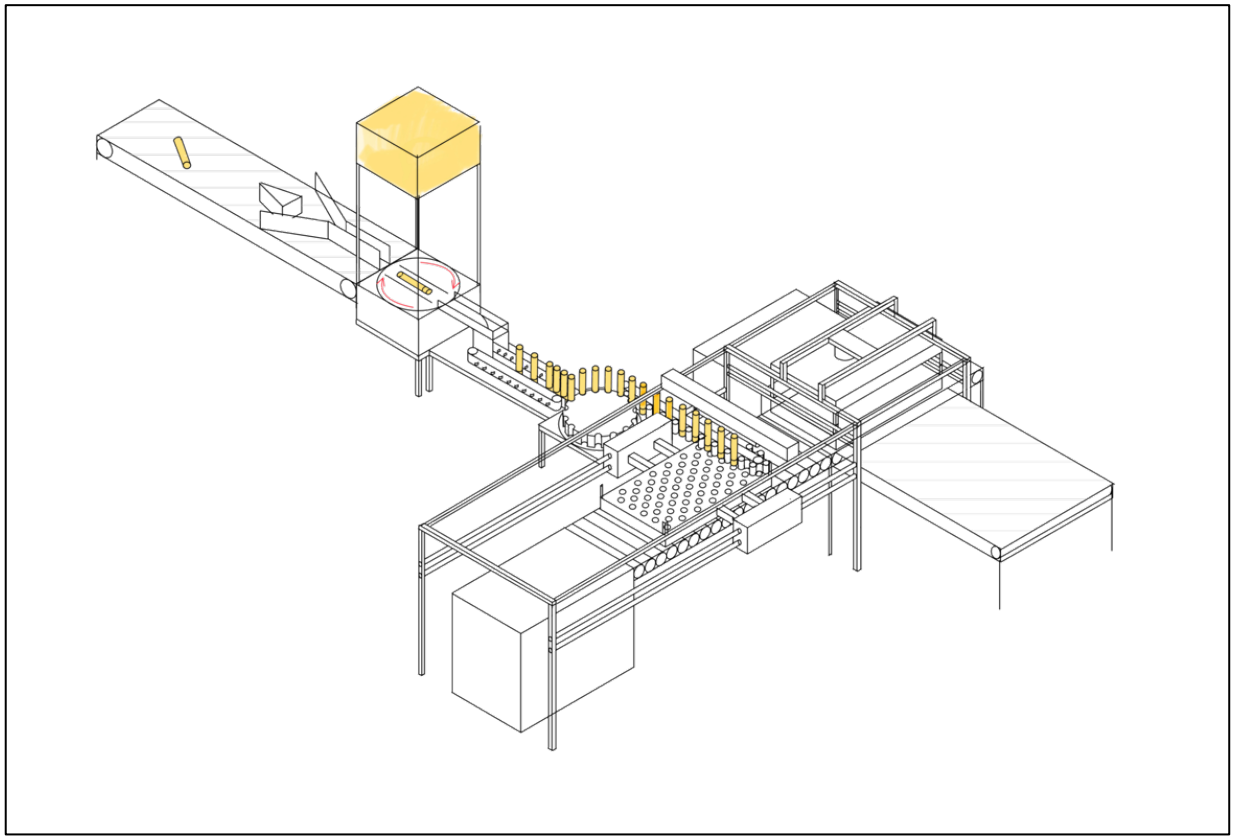


Figure 34. *Luka's Concept Sketch*

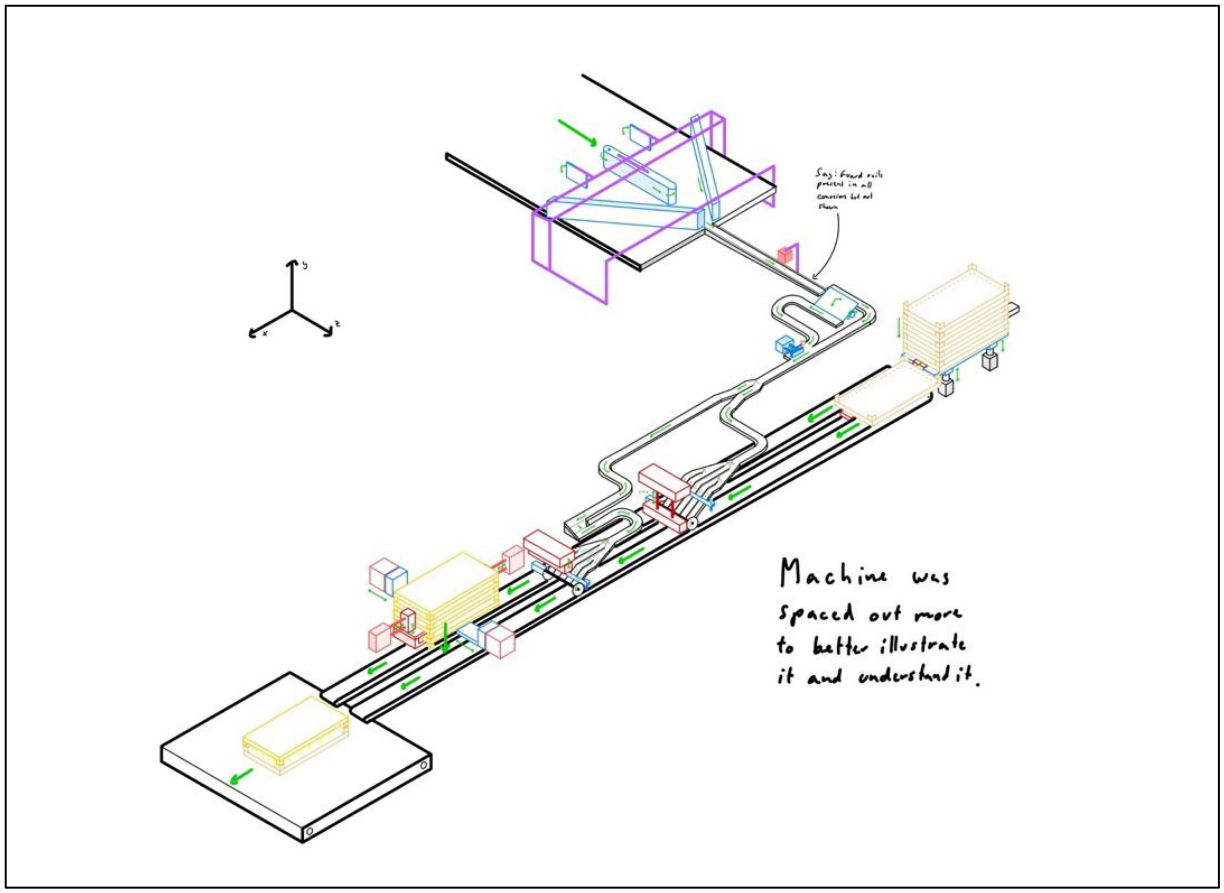


Figure 45. *Luca's Concept Sketch*

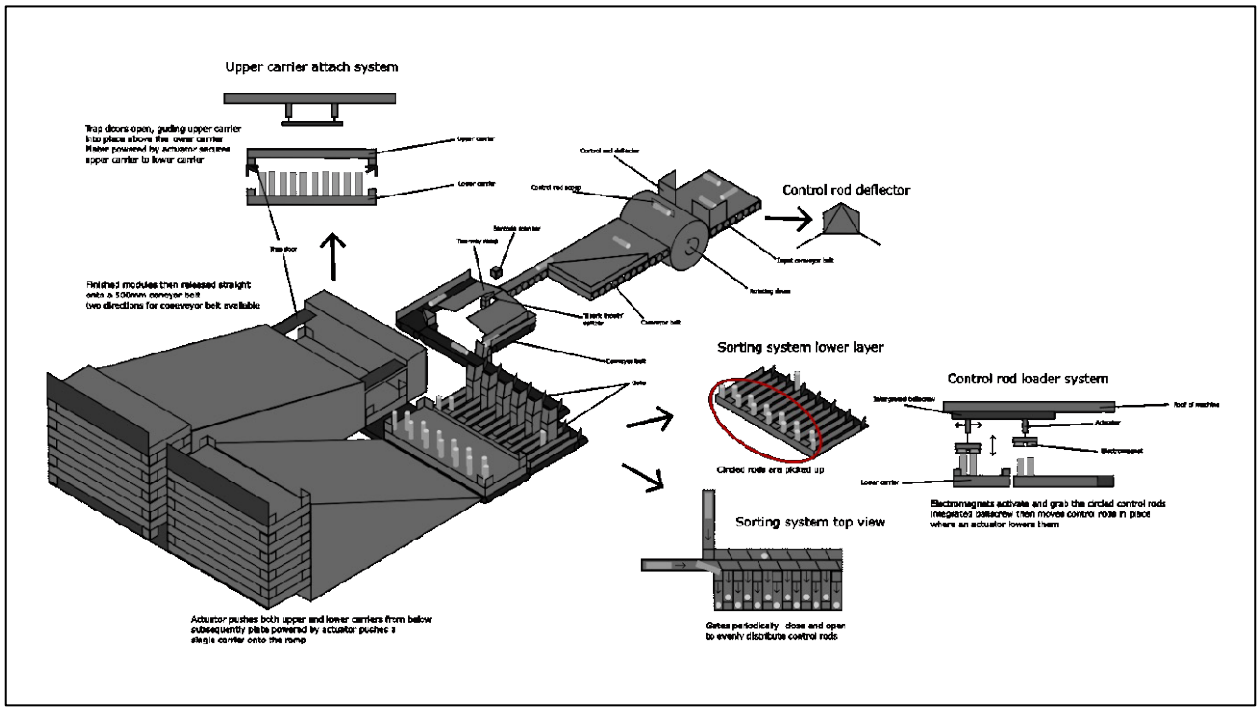


Figure 56. *William's Concept Sketch*

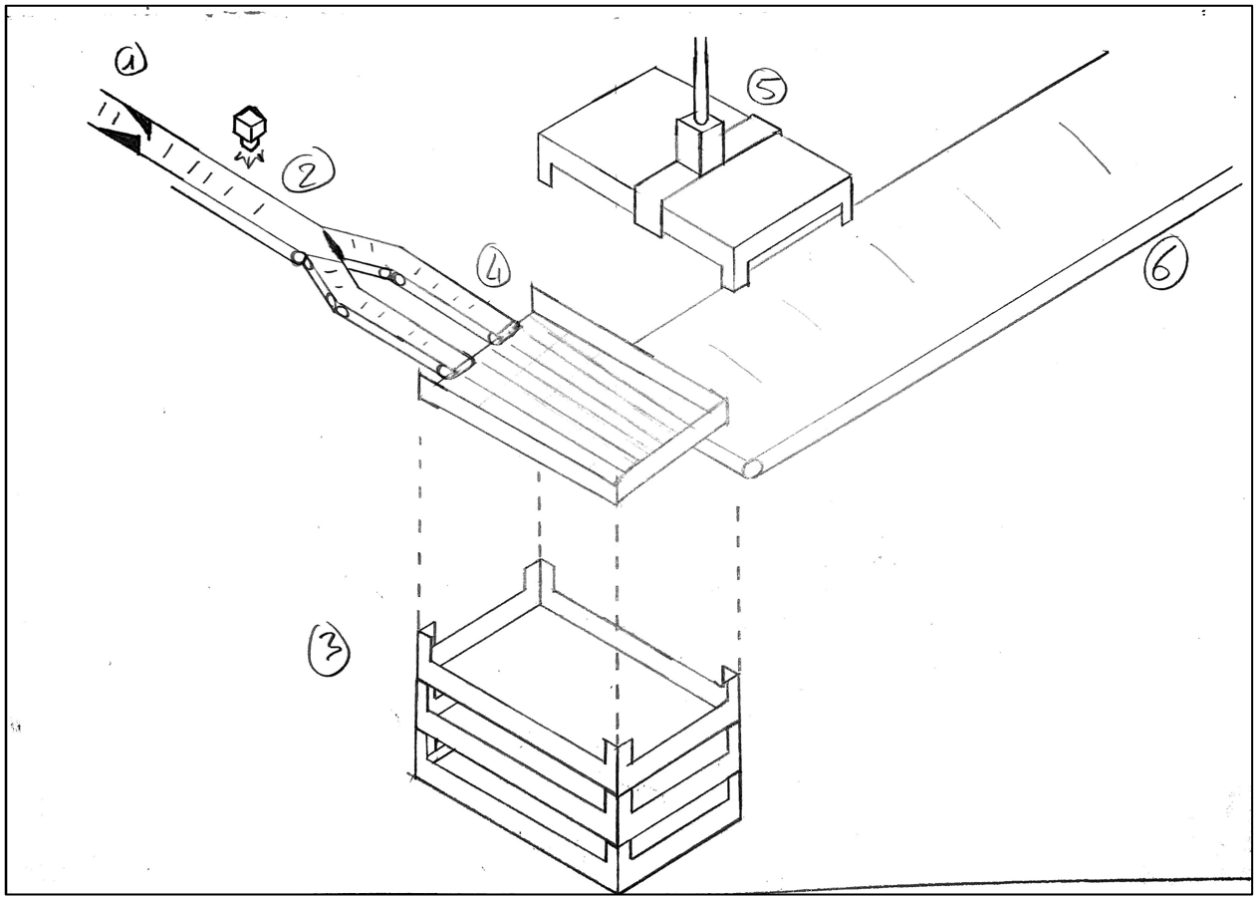


Figure 67. *Leon's Concept Sketch*

Final Machine Concept

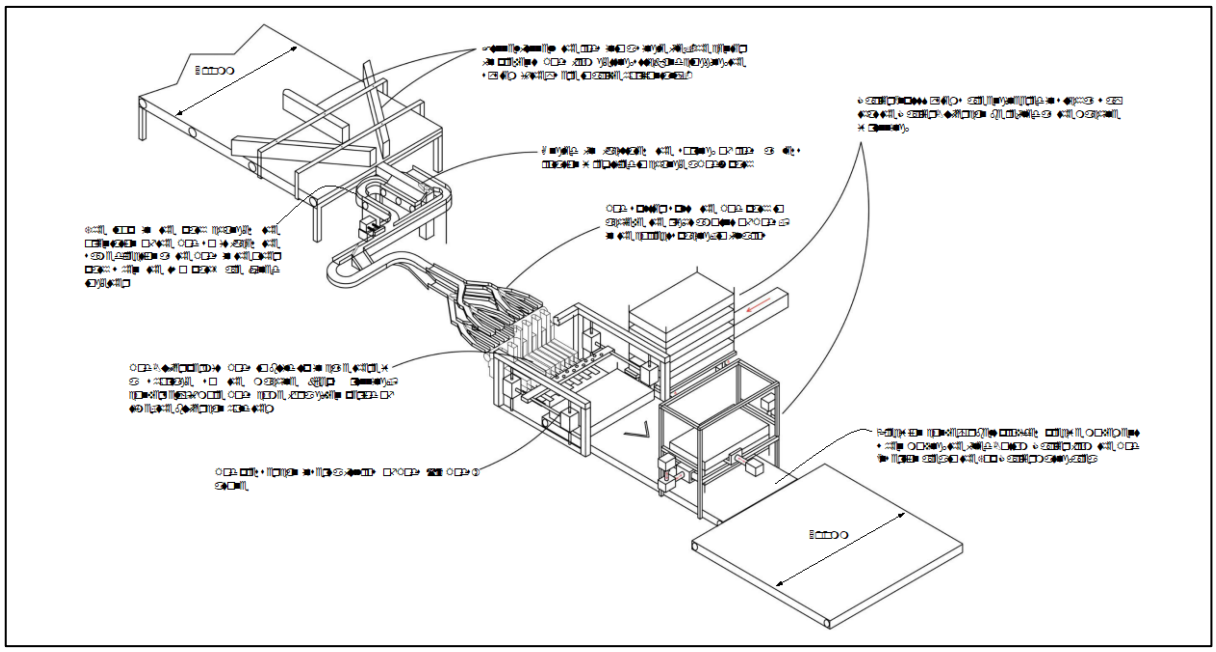


Figure 78. *3-D Sketch of Final Machine Concept*

The funnel of the machine, as seen in Figure 7, is designed to make sure no control rods get stuck, jamming the machine. Which would hinder the process of assembling a module. The rods are then scanned and separated according to their orientation; one of two paths flips the orientation of the rods to have them all face the same way. The two paths are then re-joined to negate the effect of the rod's orientation distribution not being 50-50. The rods are then split once again and go onto two levels. The top level splits the rods 6 ways whilst the bottom level splits the rods 7 ways. This totals to 13 which matches the number of rods in a row. The lower carriers are inputted onto the belt where the assembly will take place. Before reaching the magnetic arms that insert them, the rods are rotated vertically and enter a buffer, the magnetic arm then presses the rods into the Carrier a row at the time. Once the 5 rows have been inserted, the half-assembled module travels down the belt. It stops under the upper carrier input system, where the upper carrier is then pressed into the module, completing the process. Once a module is fully assembled, it is sent out on the 500 mm wide belt. There are sensors situated at various points around the machine to alert the technicians in case of any issues or component failures.

Method of Operation Storyboard

This section makes the use of high-quality CAD renders to show the operation of the machine. This will also help the technicians operating it understand the functions and how the various components are assembled. For sensor input and actuator control, refer to revised UML.

Machine Preparation

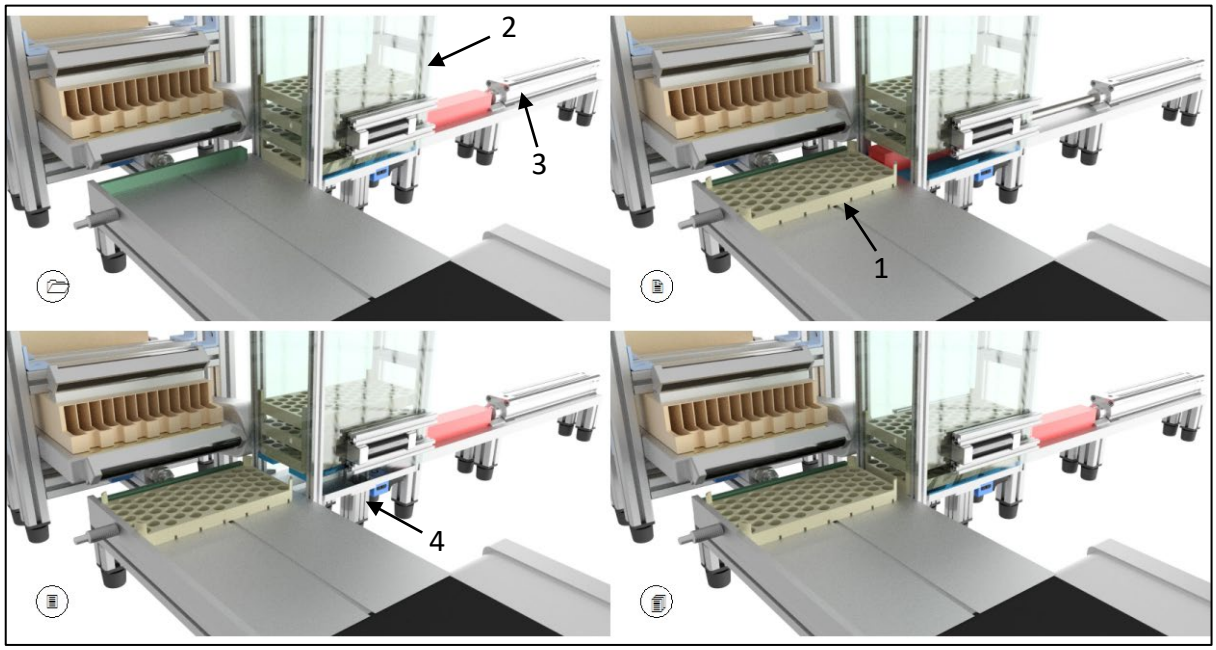


Figure 8. *Bottom Carrier Insertion (the Rod Pressing and Upper Carrier Insertion Mechanisms are hidden to ease visibility)*

Before the rods are inputted, the carriers are prepared so that they can receive the rods. A pneumatic actuator (3) pushes the bottom carrier (1) from the bottom of the feed (2) into position. The piston (red) prevents the rest of the stack from falling due to its geometry, carrying plates (blue) then lift and lower the stack once the piston has retracted.

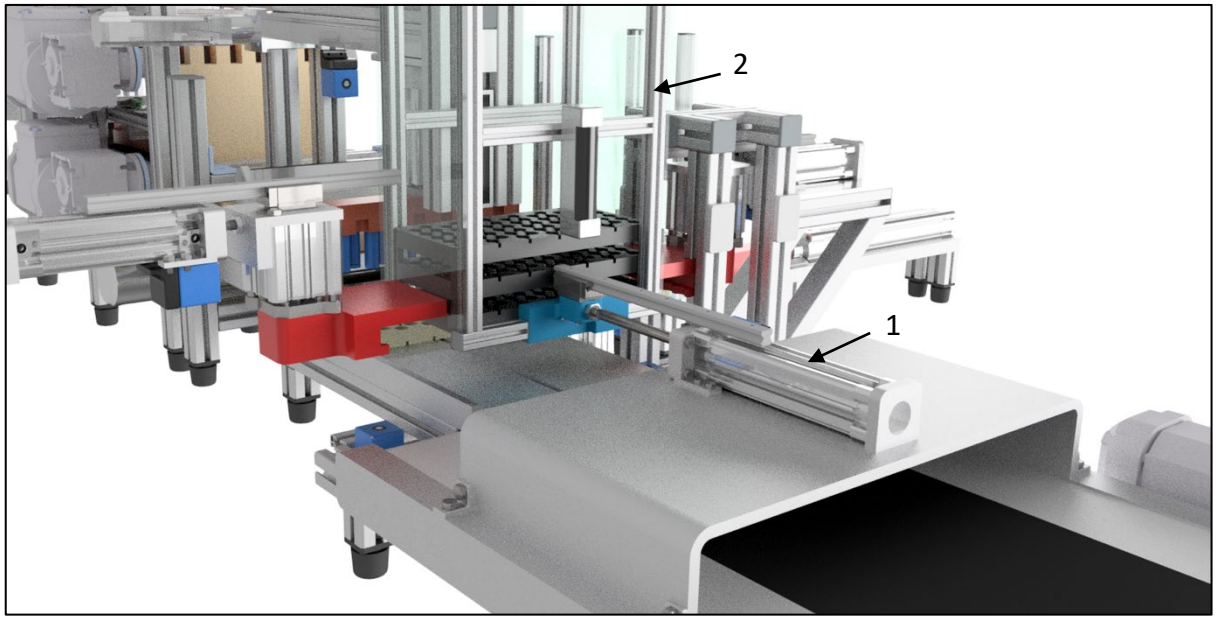


Figure 9. *Upper Carrier preparation*

The upper carrier is then put into position, the L plate (blue) is pushed forward via an actuator (1) and the upper carrier stack is then placed in the buffer (2). The maximum carrier buffer capacity is 20 carriers for both systems.

Rod Funnelling System

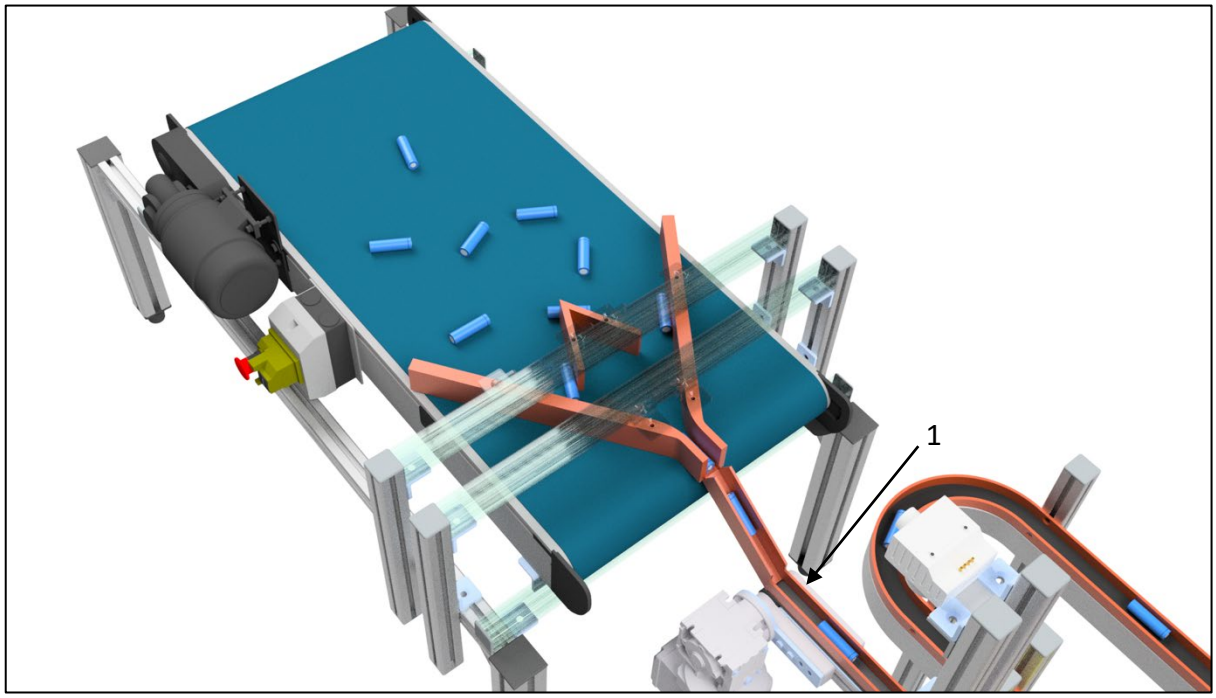


Figure 10. *Input Conveyor Belt (some beams were made transparent to ease visibility)*

Rods are guided by the central and outer funnel elements (shown in orange) to the exit passage and down a ramp towards the orientation scanning sub-assembly conveyor (1).

Rod Scanning, Sorting and Splitting

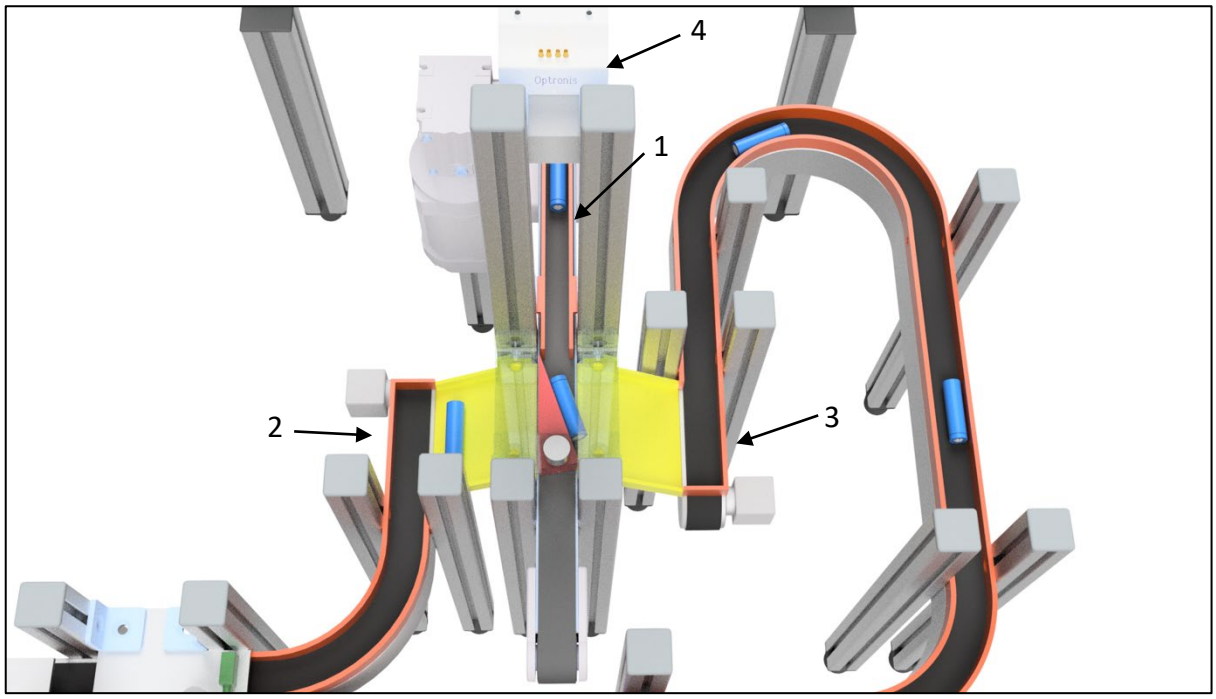


Figure 11. *Rod orientation scanning and initial splitting (covers were made transparent to ease visibility)*

Rods enter on conveyor (1) under the camera (4), which scans the orientation. Depending on the orientation rods are split left or right onto conveyors (2) and (3), conveyor (3) rotates the rods 180 degrees to be the same orientation as the rods is conveyor (2).

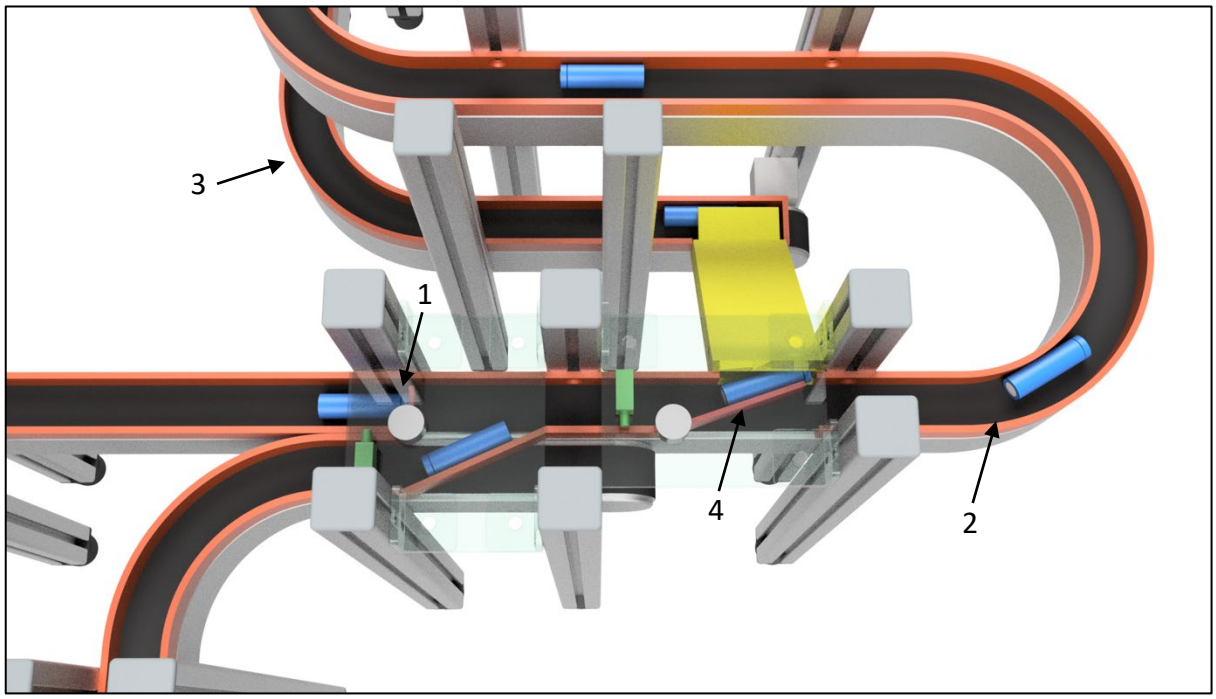


Figure 12. *Rod re-joining and subsequent splitting (covers were made transparent to ease visibility)*

Rods, which now face the same direction (moving left to right), are re-joined into the same conveyor using a sensor (green) and blocker (1). Rods are then split 6-7 to the upper (2) and lower (3) conveyors respectively by a splitter fin (4).

Rod Splitting and Alignment System

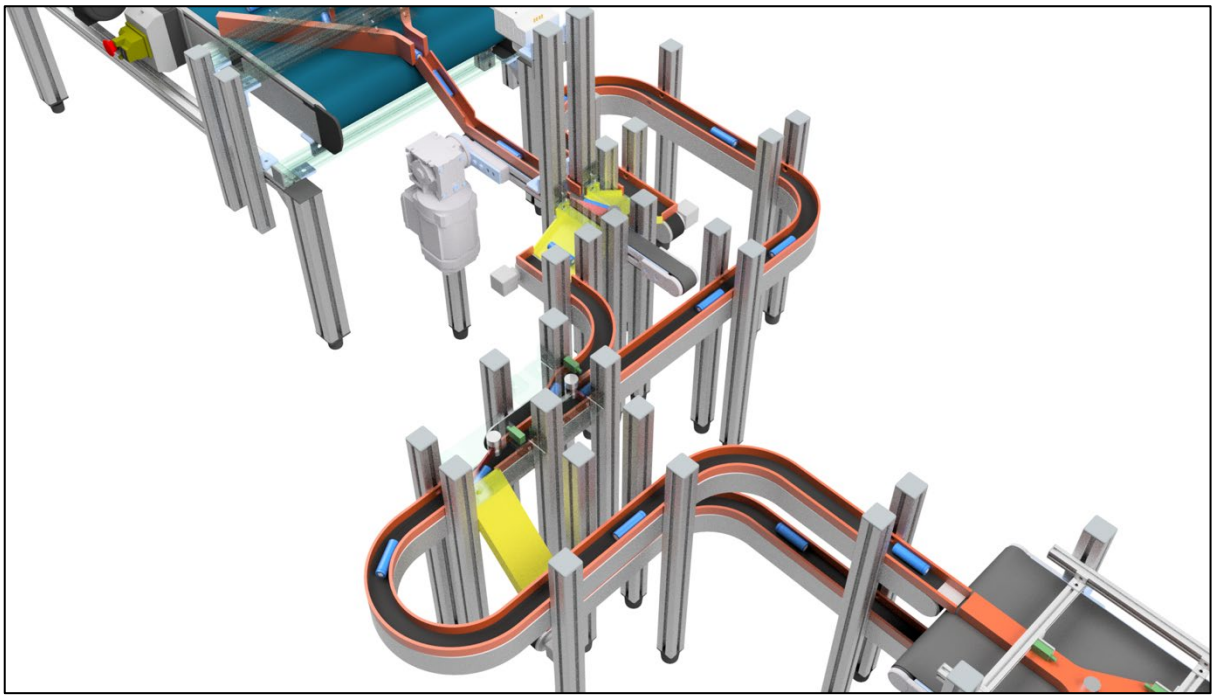


Figure 13. *Rod scanning and splitting arrangement shown for context (covers were made transparent to ease visibility)*

After having been sorted, oriented, and split, the rods make their way to the 6- and 7-way splitting channels that sit on top of conveyor belts.

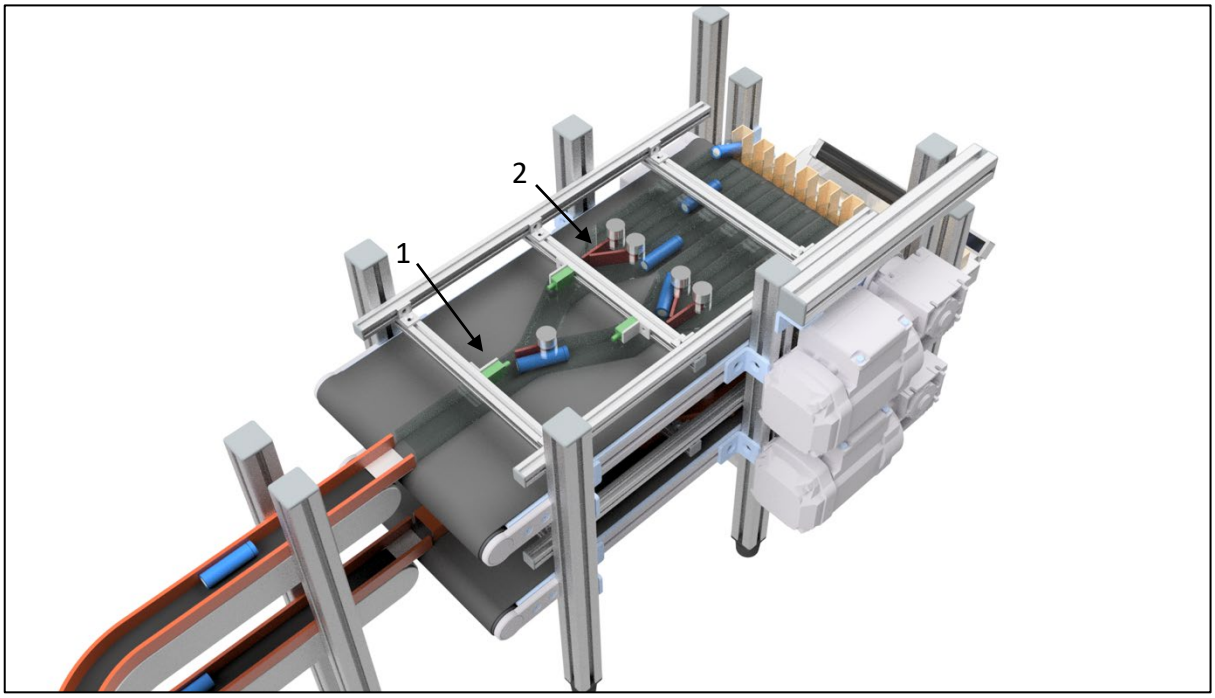


Figure 14. *6-way Rod Splitter (the splitter made transparent to ease visibility)*

Rods on upper and lower curved conveyors, shown with orange rails, are pushed into the splitting channels, made transparent for visibility. The rods are then evenly split between 6 and seven channels using the sensors (1) and fins (2).

Rod Buffer System

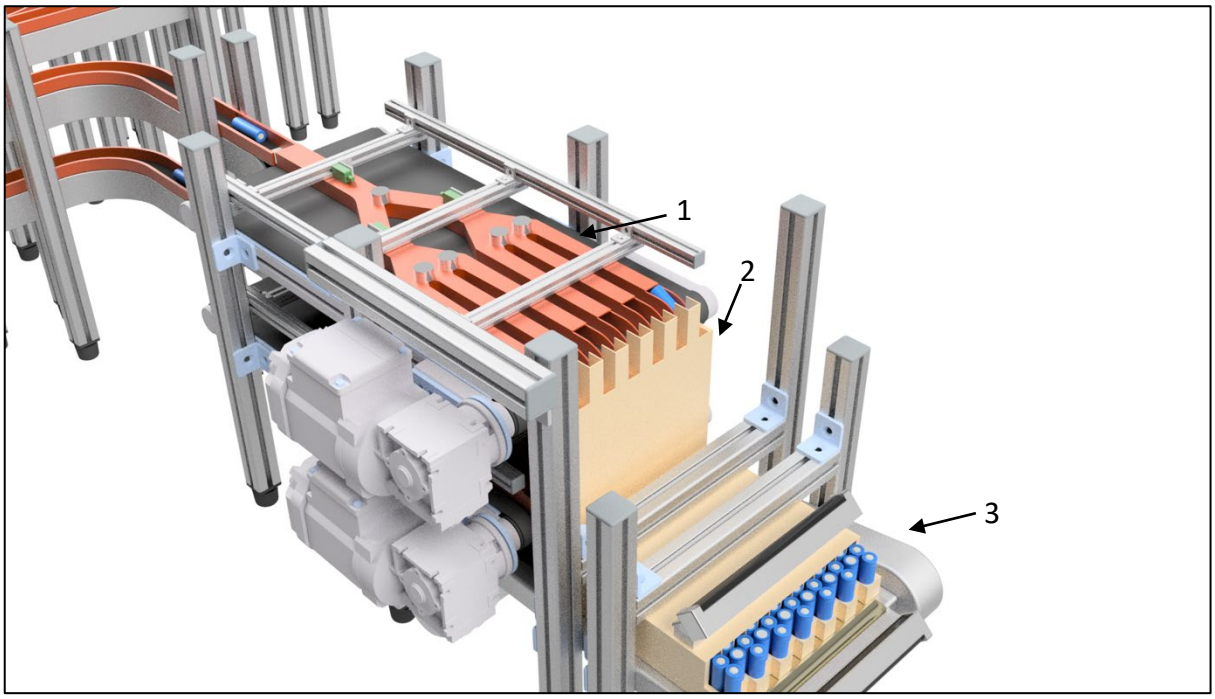


Figure 15. *Rod Buffer (Carrier Assembly and Output System is hidden to ease visibility)*

The rods then drop from the channels (1) into buffer (2) where a magnetic conveyor belt (3) pushes them forward and keeps them upright. When two or more rows are filled, the light gate signals for rod insertion to begin.

Rod Extraction

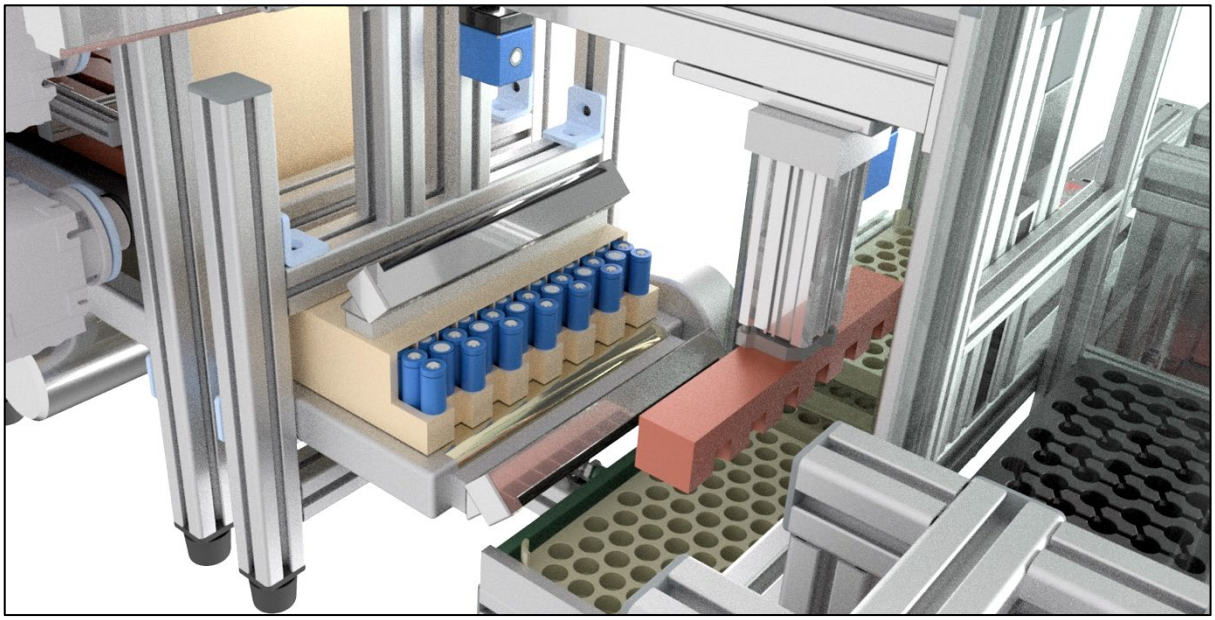


Figure 16. *Rod Buffer at full capacity and Rod Presser preparing to move and grab the Rods*

The rod buffer, whose geometry matches the pattern of the carrier, allows for a horizontal row of rods to be inserted into the carrier at once.

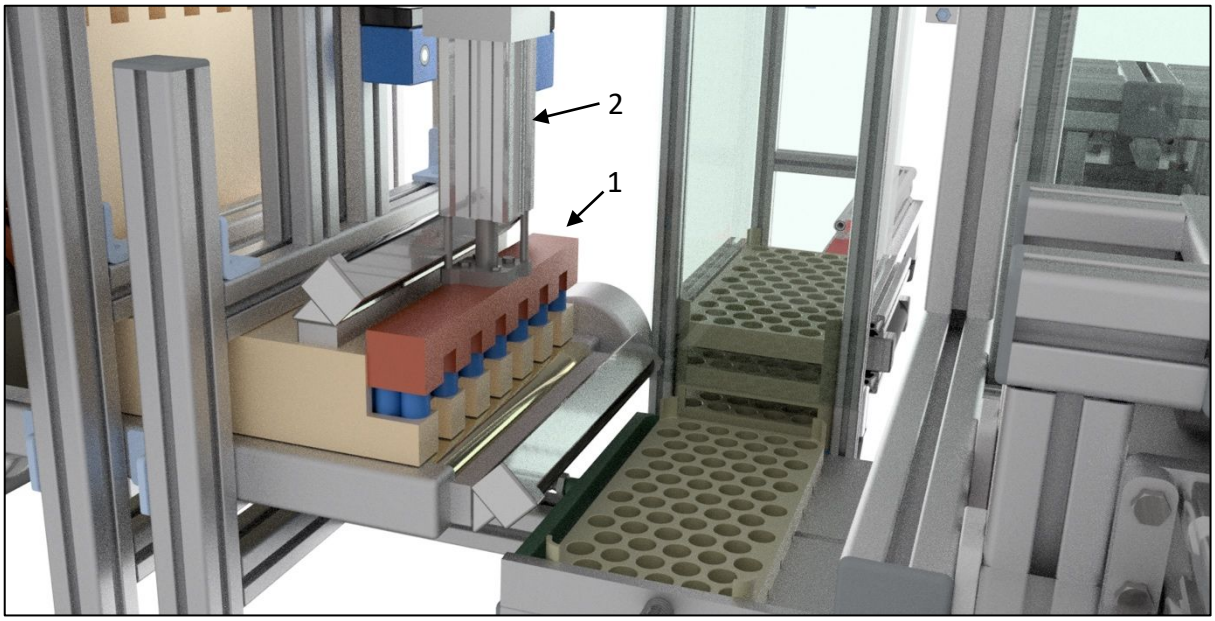


Figure 179. *Rod Presser grabbing the Rods*

When a row of rods is full, the Presser (1) is moved over the buffer by a linear unit. The presser actuator (2) then extends, and the magnets located inside the presser pick up the front row of rods.

Rod Insertion

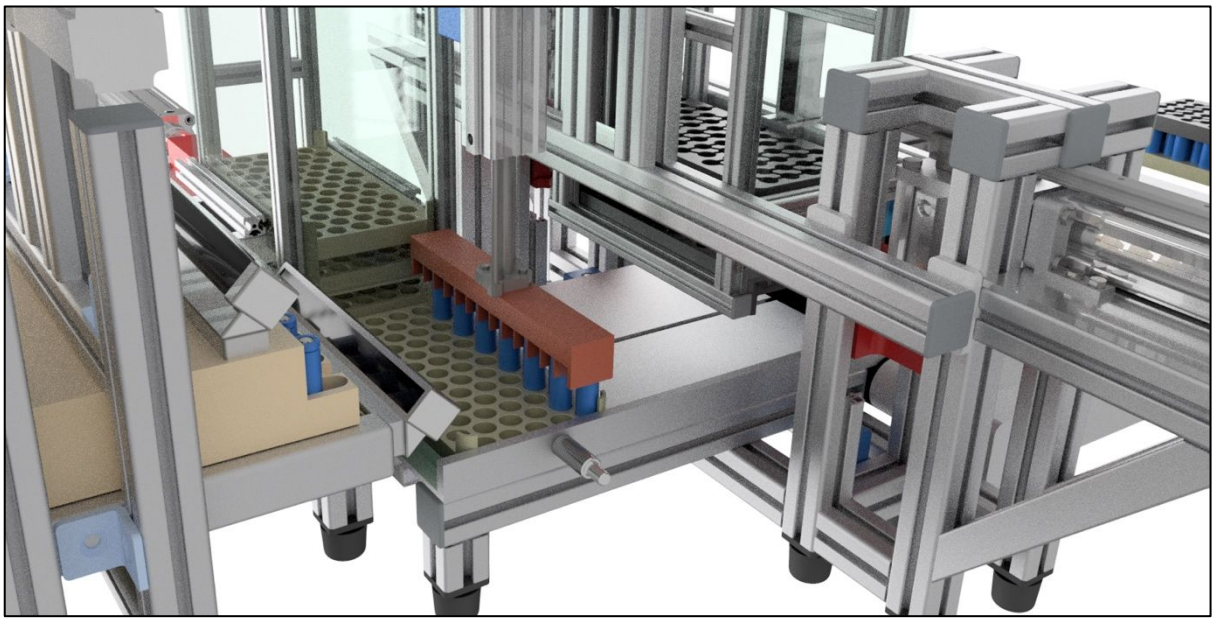


Figure 18. *Rod Presser pressing the first row of Rods into the Bottom Carrier*

The presser actuator then retracts, moves into position, and presses the first row of rods into the bottom carrier with a force of 650 N. The actuator then retracts.

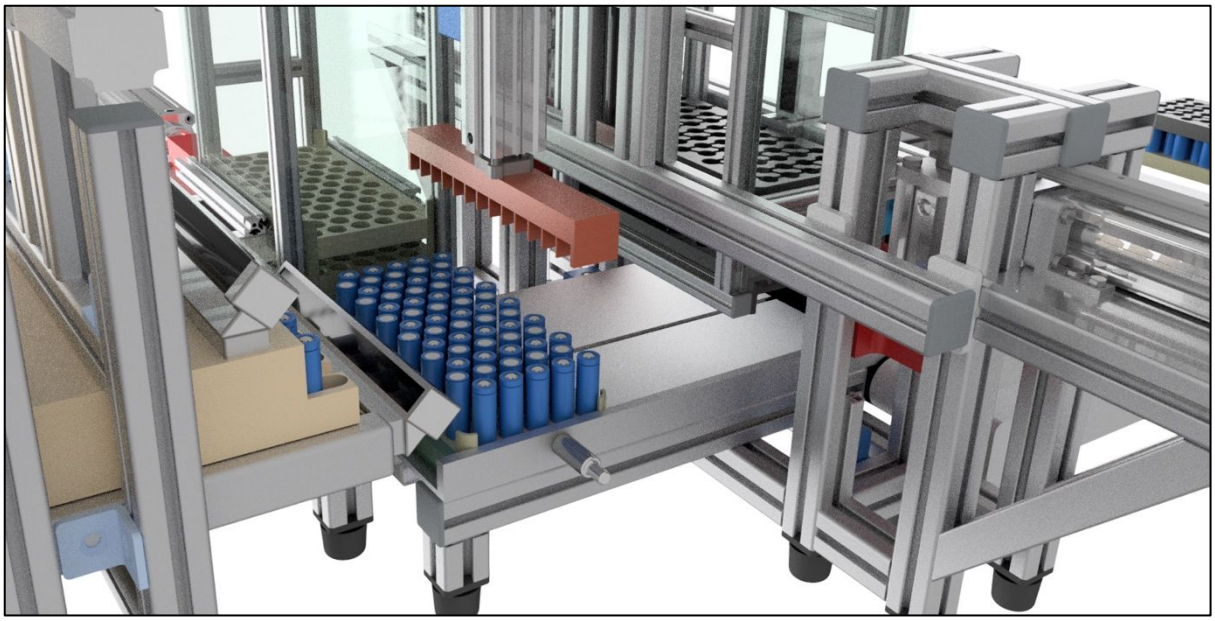


Figure 19. *Bottom Carrier filled with Rods*

The previous process is repeated five times, moving the bottom carrier forward slightly each time to position the presser above the next row, until the bottom carrier is full.

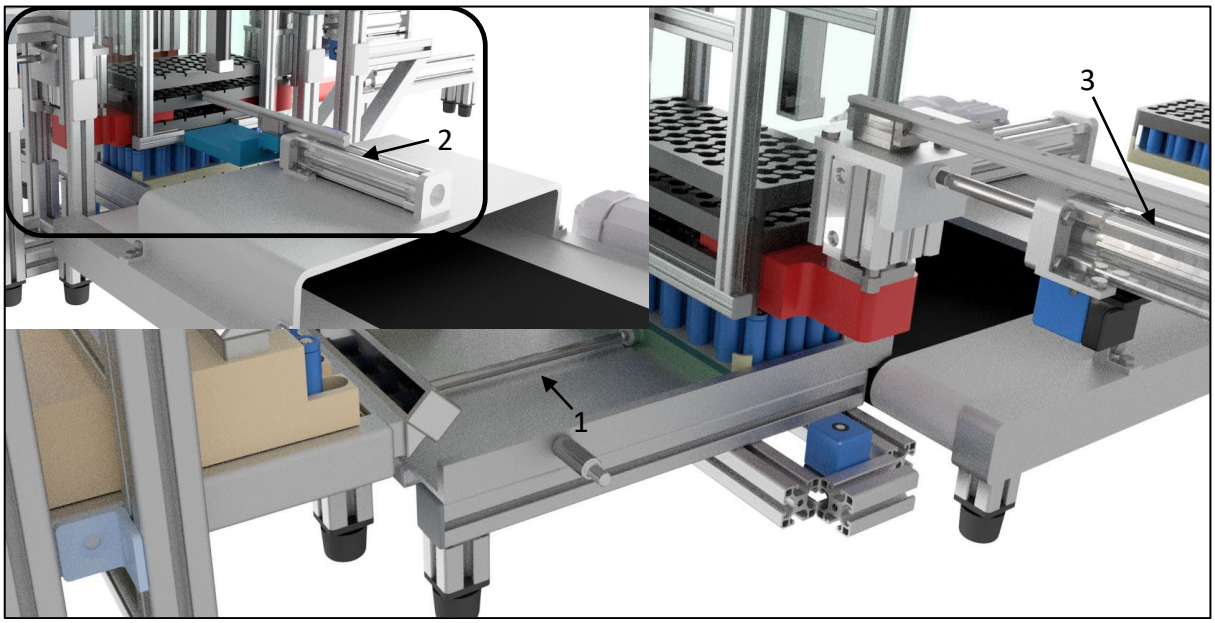


Figure 20. *Filled Bottom Carrier pushed in position by an actuator (Red and Blue Support Structures hidden to ease visibility)*

A pneumatic actuator (1) pushes the half-complete module under the top carrier input system. The L plate (shown in blue) is retracted by an actuator (2) and the two clamps (shown in red) are then moved in by an actuator (3), clamping the top carrier.

Final Module Assembly and Output

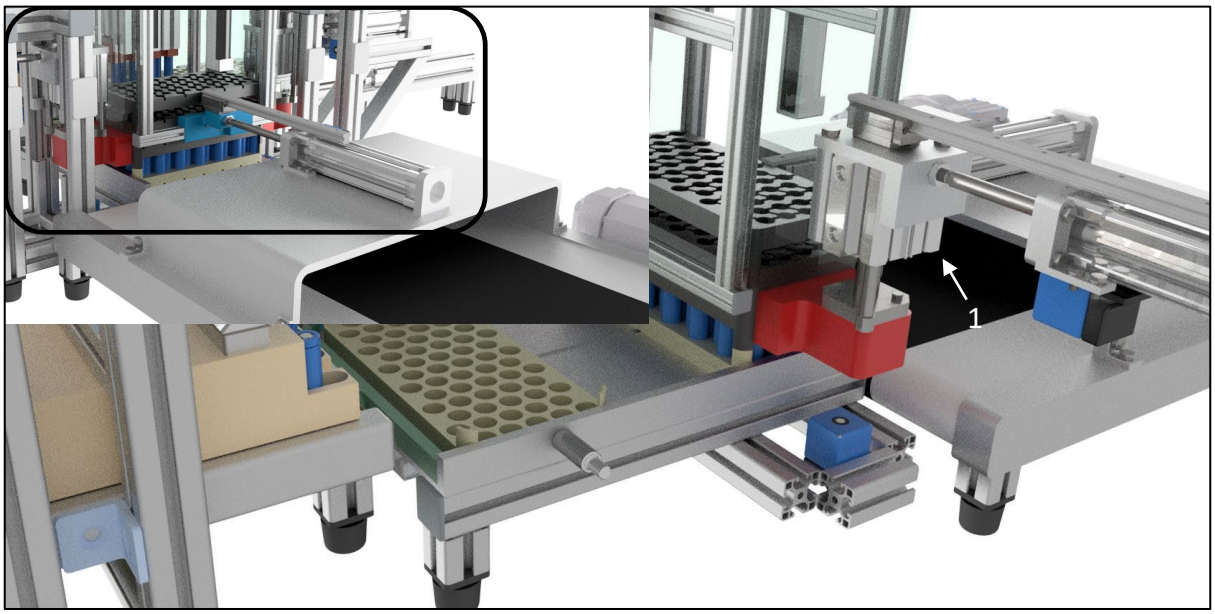


Figure 21. *Carrier mating / Final Module assembly (Red and Blue Support Structures hidden to ease visibility)*

The upper carrier stack is then lowered by the clamps and actuators, delivering 1,625 N of downwards force each (1); this mates the lowest carrier in the stack with the filled bottom carrier. The L plate then is moved back in to support the stack of upper carriers and the clamps retract and move up to be reset to their starting position.

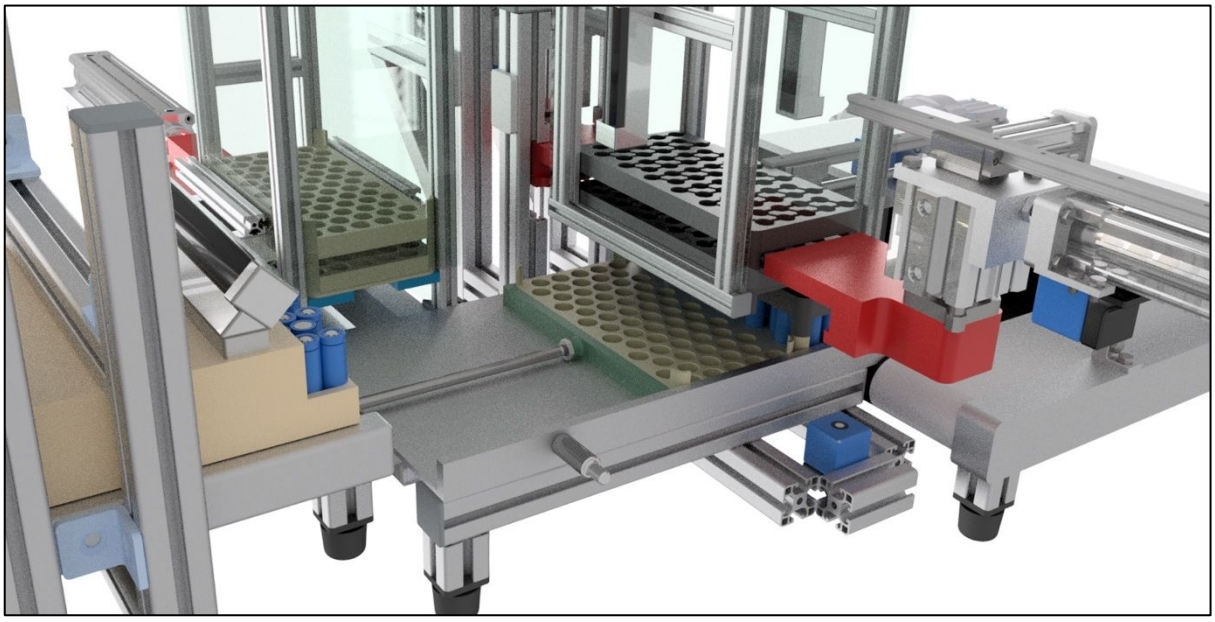


Figure 22. *Assembled Module being pushed out by a filled Bottom Carrier (Red and Blue Support Structures and Rods in Carrier hidden to ease visibility)*

After another lower carrier is filled with rods, it is pushed by a pneumatic actuator to where it will be mated with the upper carrier. Doing so pushes the previously completed module onto the output conveyor belt.

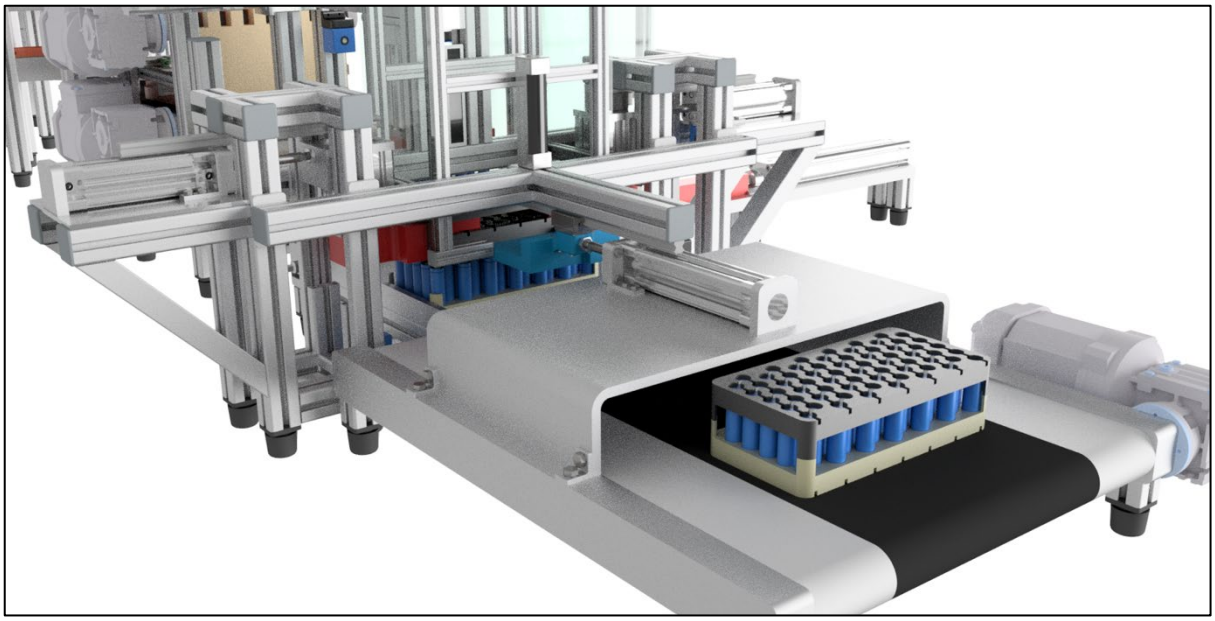
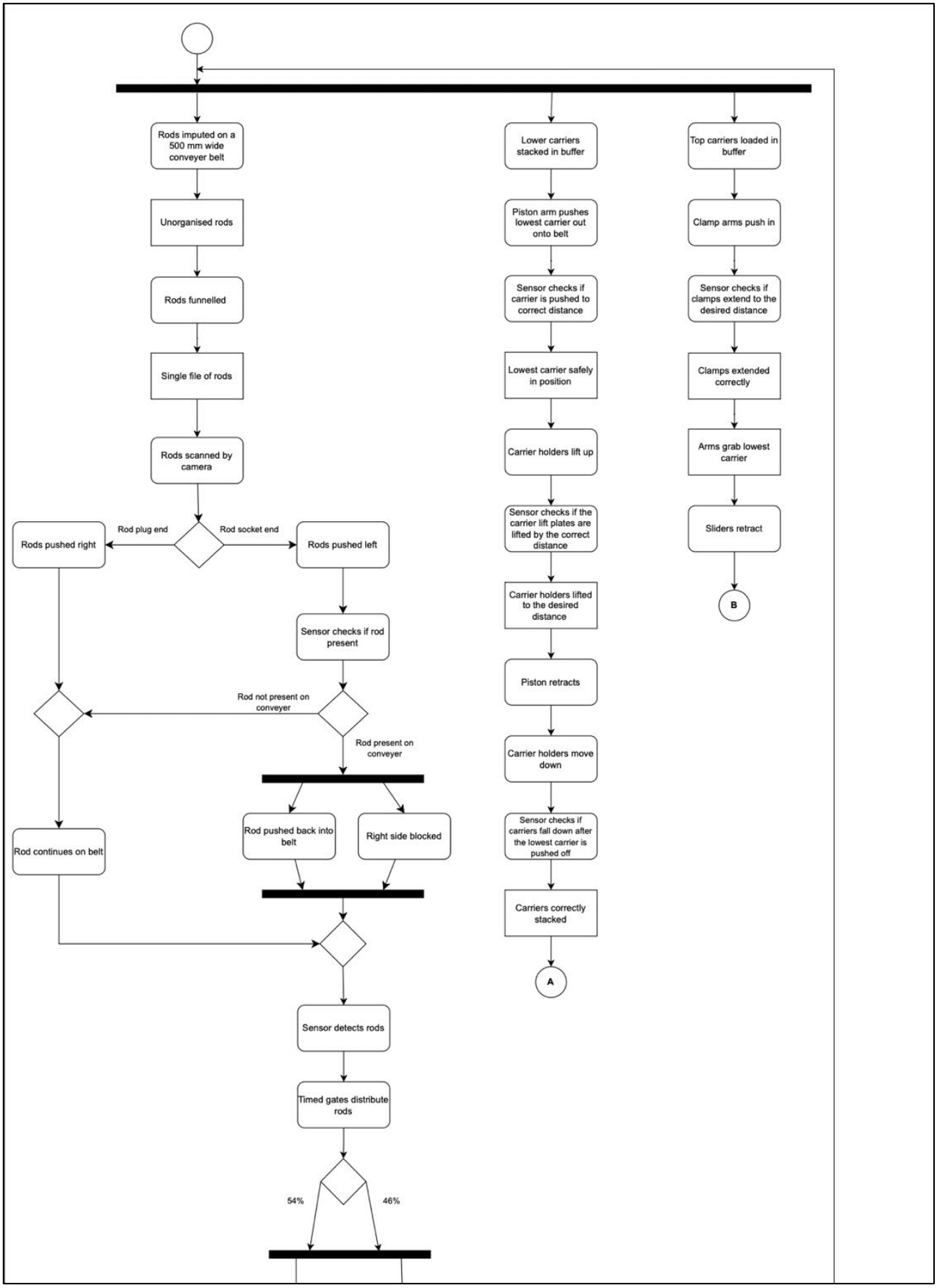
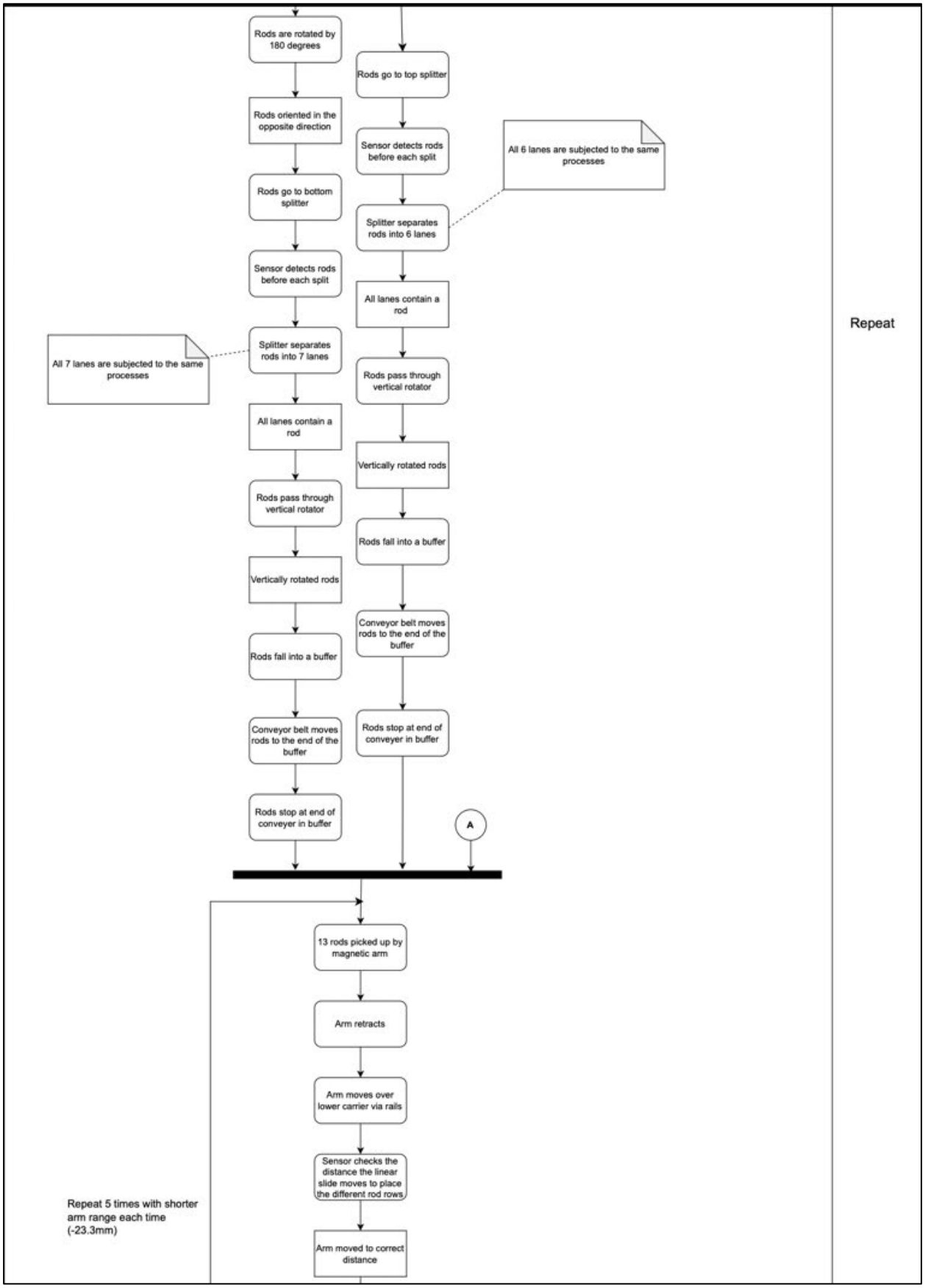


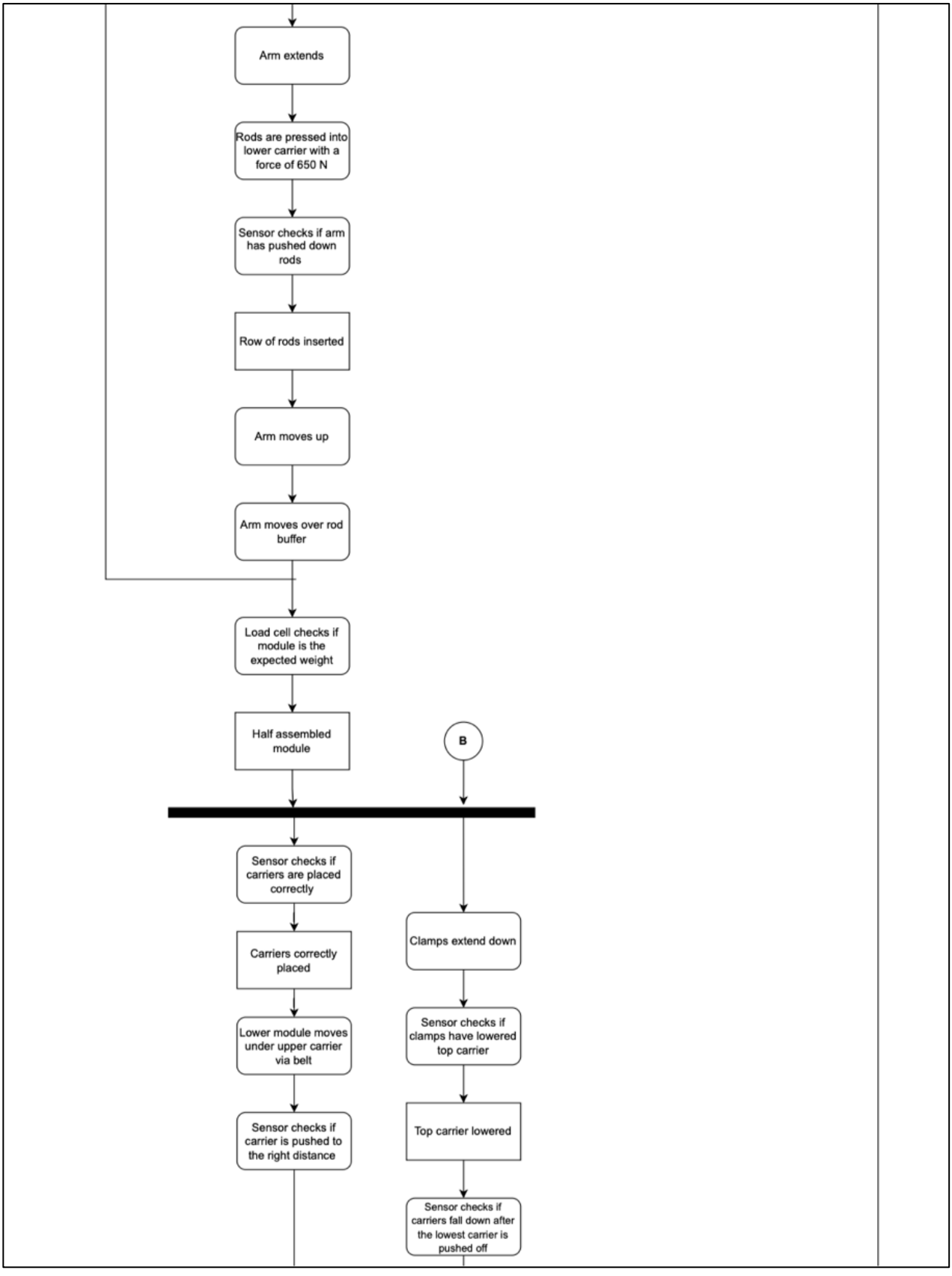
Figure 23. *Assembled Module being carried by the Output Conveyor Belt to its next destination*

The completed module is then transported to its destination via the output conveyor belt.

UML Activity Diagram




Rods are rotated by 180 degrees
Rods oriented in the opposite direction
Rods go to bottom splitter
Sensor detects rods before each split
Splitter separates rods into 7 lanes
All 7 lanes are subjected to the same processes
All lanes contain a rod
Rods pass through vertical rotator
Vertically rotated rods
Rods fall into a buffer
Conveyor belt moves rods to the end of the buffer
Rods stop at end of conveyer in buffer
Rods go to top splitter
Sensor detects rods before each split
Splitter separates rods into 6 lanes
All 6 lanes are subjected to the same processes
All lanes contain a rod
Rods pass through vertical rotator
Vertically rotated rods
Rods fall into a buffer
Conveyor belt moves rods to the end of the buffer
Rods stop at end of conveyer in buffer
A
13 rods picked up by magnetic arm
Arm retracts
Arm moves over lower carrier via rails
Sensor checks the distance the linear slide moves to place the different rod rows
Arm moved to correct distance
Repeat 5 times with shorter arm range each time (-23.3mm)
Repeat


Arm extends
Rods are pressed into lower carrier with a force of 650 N
Sensor checks if arm has pushed down rods
Row of rods inserted
Arm moves up
Arm moves over rod buffer
Load cell checks if module is the expected weight
Half assembled module
B
Sensor checks if carriers are placed correctly
Carriers correctly placed
Lower module moves under upper carrier via belt
Sensor checks if carrier is pushed to the right distance
Clamps extend down
Sensor checks if clamps have lowered top carrier
Top carrier lowered
Sensor checks if carriers fall down after the lowest carrier is pushed off

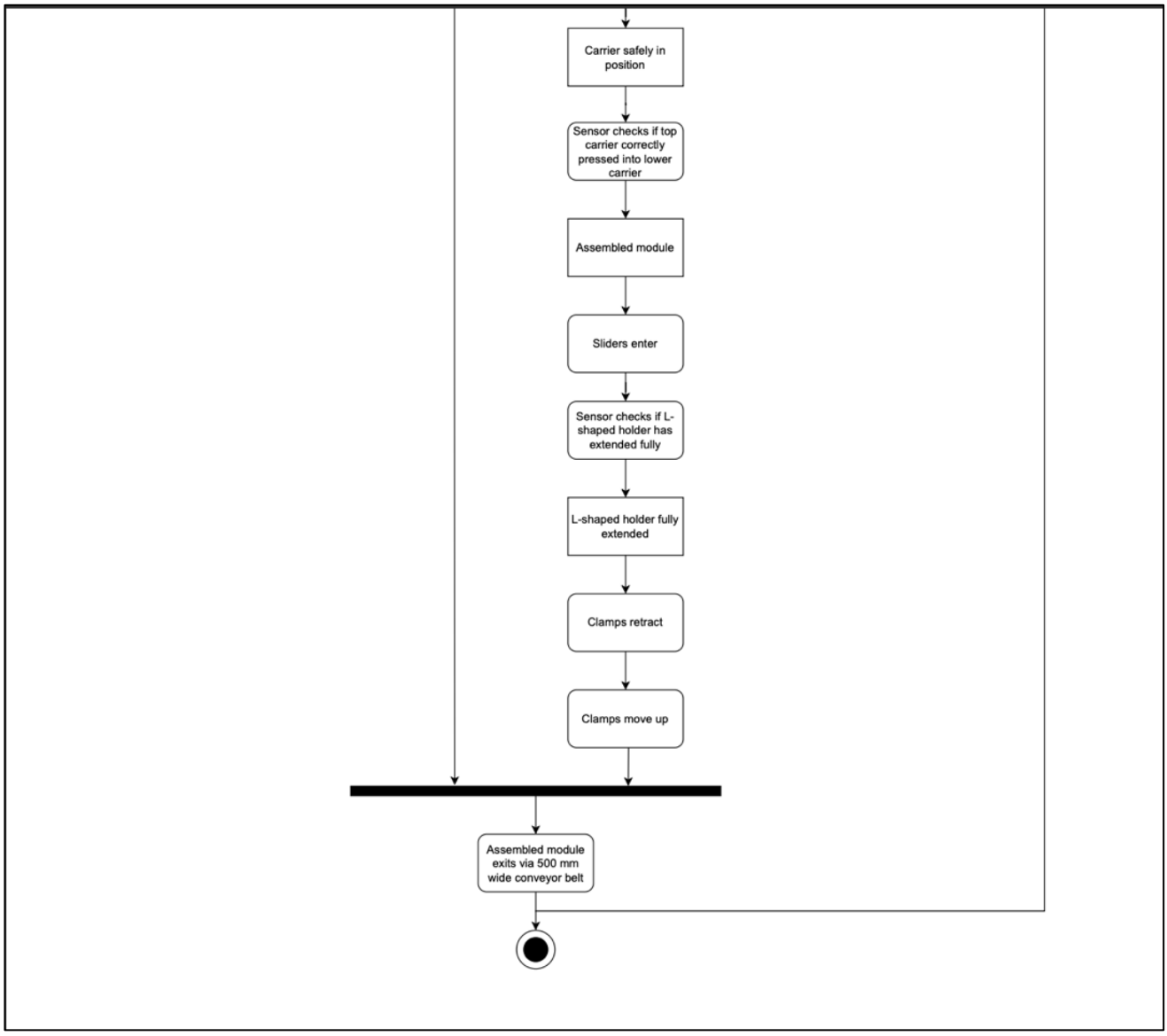


Figure 24. *UML Analysis Diagram of Machine*

The UML diagram details the process of module assembly, from the input of rods to the output of the assembled modules, including the design's architecture, the sensors used and control mechanisms. When we made the UML Activity Diagram of our final machine, we decided to separate the assembly into three parts. The first part details the path that the rods follow during the entire process. The second part that added **(A)**, shows the input of bottom carriers and the process of setting them up for assembly. The last section **(B)** we added is the input of upper carriers and their preparation for assembly. They all combine again at different parts during the UML and come out as a fully assembled final product.

FMECA Table and Fault Tree Diagram

Upon conducting failure modes, effects, and criticality analysis for the machine sub-assembly responsible for assembling the module, we determined that the predominant risk factor revolves around the multitude of actuators employed in this phase. These components play a crucial role as they constitute the primary elements involved in the assembly of a module, which is the core objective of this machine. Notably, our least vulnerable components are the magnets and the linear transmission unit, whose reliability hinges on rare external factors.

For the machine sub-assembly responsible for setting up the rods before insertion (the rod splitter and buffer), we used a fault tree analysis diagram. This allowed us to observe the possible sequence of events that could lead to this process going wrong. Most failure modes we found are avoidable thanks to the design we have come up with. This should limit the possibility of this sub-assembly failing.

FMECA									
Sub-Assembly: Module Assembly									
Part Name	**Function**	**Failure Mode**	**Cause**	**Effect**	**Rating out of 10**			**RPN**	**Failure Elimination/Reduction**
					Severity	Occurrence	Detection		
Pushing Actuator (20 mm bore, 300 mm stroke)	Pushes filled lower carrier from where it recieves the rods, to where it is mated with the upper carrier	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, incorrect procedures	Lower carrier stack not alligned with upper carriers, preventing module completion	**8**	**3**	**4**	96	Ensure the equipment is correctly maintained, and that it is used for the right processes
Linear Transmission Unit	Translate rods and presser from buffer to lower carrier	Does not move to the correct positions when actuated	Power cut, damage of the parts, error in code	The rods will not be pressed in the correct places leading to damage in the pressing actuator and rods	**9**	**3**	**1**	27	Ensure continous current supply, dry operating conditions, sensor input to check operation
Magnets	Picking up and carrying rods from buffer over to lower carriers	Magnets not stron enough to pick the rods up	Magnets get de-magnetised with use	Rods will not be trasnported from the buffer to the lower carriers	**9**	**2**	**1**	18	Ensure the magnets are used in a suitable environment, free of any damaging conditions
Pressing Actuator (40 mm bore, 100 mm stroke)	Press rods in lower carrier	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, incorrect procedures	Rods not inserted into carrier, preventing module completion	**9**	**4**	**3**	108	Ensure the equipment is correctly maintained and that it is used for the right processes
Pneumatic Actuator (25 mm bore, 40 mm stroke)	Supports the lower carriers in their buffer	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, incorrect procedures	Could cause damage to upper carriers and delay module completion process	**7**	**4**	**3**	84	Ensure the equipment is correctly maintained and that it is used for the right processes
Pneumatic Actuator (63 mm bore, 40 mm stroke)	Pushes upper carrier into lower carrier to complete module	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, incorrect procedures	Will prevent completion of the modules	**9**	**4**	**3**	108	Ensure the equipment is correctly maintained and that it is used for the right processes
Pneumatic Actuator (32 mm bore, 85 mm stroke)	Pushes the L plate to support the other upper carriers in the buffer	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, inserting to many carriers	Subsequent upper carries can not be connected to lower carrier stack	**7**	**4**	**3**	84	Ensure the equipment is correctly maintained and that it is used for the right processes
Pneumatic Actuator (50 mm bore, 300 mm stroke)	Pushes lower carrier into position to recieve the rods	Excessive loading, exceeding duty cycle, contamination	Poor maintenance of equipment, incorrect procedures	Will slow down/stop process of assembly if no lower carrier is available	**8**	**4**	**3**	96	Ensure the equipment is correctly maintained and that it is used for the right processes
Clamp	Clamping carriers for further transportation and mating of module	Clamp can fail due to rust and corrosion or due to eccessive loading	Poor maintenance of equipment, inadequate environment	Will stop upper carriers from being carried and connected to lower carriers	**8**	**4**	**3**	96	Ensure the clamps are correctly maintained and used in adequate condiitons
L-shaped holder	Maintaining buffer of upper carriers in place	Failure due to eccessive loading	Errosion due to poor maintenance , overload, fatigue	Will cause upper carriers in the buffer to collapse onto the belt causing possible damage and impacting assembly process	**7**	**3**	**3**	63	Ensure the component is correctly maintained and can withstand the loads

Severity(S): 1 - 10	Occurrence(O): 1- 10	Detection(D): 1 - 10	RPN (Risk Priority Number) = S x O x D
1: not severe	1: Unlikely	1: Easy to detect	
10: Maximum severity	10: Maximum probability	10: Undetectable	

Figure 25. *Failure Modes, Effects, and Criticality Analysis Table*

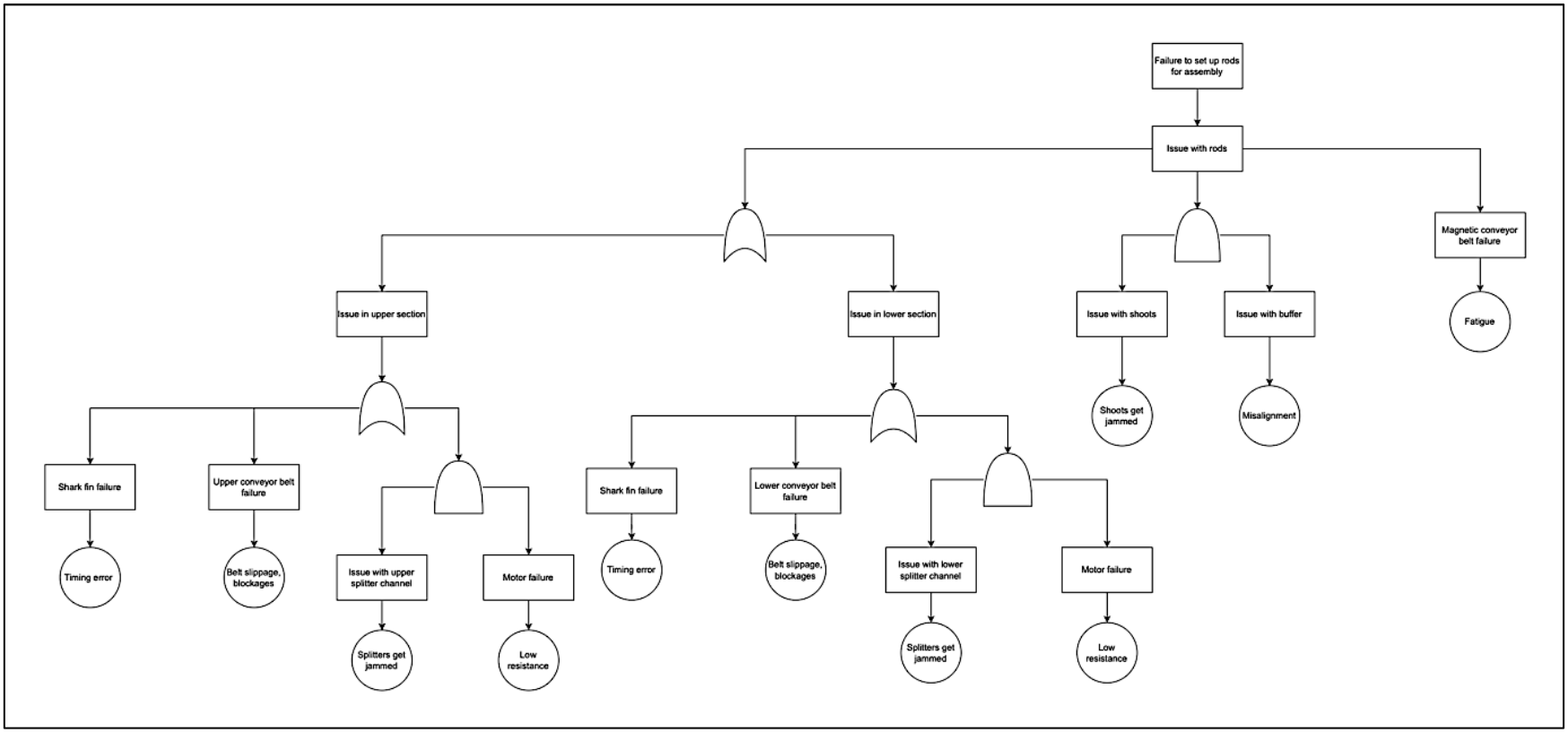


Figure 26. *Fault Tree Analysis Diagram*

Motivation for Calculations

In the retrieval mechanism for the bottom carrier, it was important to determine the maximum number of carriers achievable in the feed. More carriers would mean that the workload on operators would be reduced. This is because the carrier feeds would have to be resupplied less throughout a working day, which would reduce labour costs to operate the machine. However, too many could make handling the carriers when inserting them difficult. Furthermore, too many carriers could overwhelm the pneumatic actuators used to lift and push the carriers, and as seen in the FMECA previously, a failure at this stage in the assembly process would prevent the completion of further modules completely.

For the pneumatic actuators in the lifting load case, a 2-D force diagram was produced to evaluate the total number of carriers that could be supported. These calculations are shown in Appendix B. As seen in the load case shown in Figure 27, the primary contributor to the load on the pneumatic actuators was the weight (W) of the lower carriers. The Force P is the force of each pneumatic actuator acting to lift the carriers.

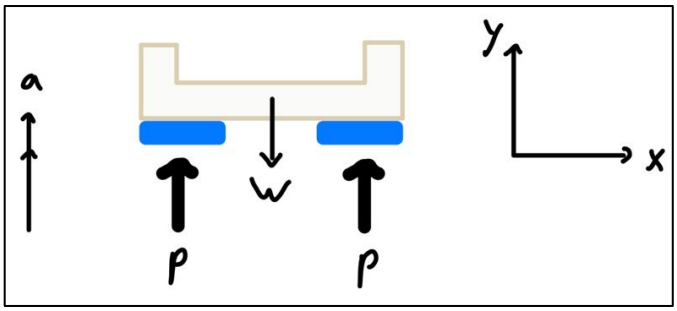


***Figure 27**. Load case for lifting pneumatic actuator*

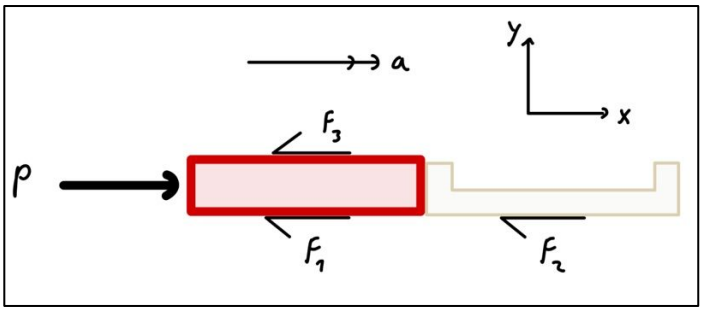


***Figure 28.** Load case for pushing pneumatic actuator*

By setting W to the weight of 100 carriers, a pneumatic actuator was selected that met the required output force while operating at 6 Bar, as required by the client. The friction acting on the pneumatic actuator for the second load case was then evaluated using the same method. These calculations are also shown in Appendix B. The friction was split into 3 components allowing friction on all surfaces to be evaluated. This is shown in the load case in Figure 28 above. Using 100 carriers it was determined that the friction was too significant. Hence, a reduction to 20 carriers in the feed was done. This allowed the less powerful and more affordable SMC CDQM Compact Cylinder Guide Rod with a 25 mm bore, and 40 mm stroke to be selected for the lifting load case. Figure 29 below shows the theoretical output of the pneumatic actuator. A Festo DSBC-50-300-PPVA-A-N3 Standard cylinder with 50 mm bore and 300 mm stroke was selected for the pushing load case. Figure 30 below shows the theoretical output of this pneumatic actuator.

Theoretical Output

Unit: N

Bore size (mm)	Operating direction	Operating pressure (MPa)		
		0.3	0.5	0.7
12	IN	25	42	59
	OUT	34	57	79
16	IN	45	75	106
	OUT	60	101	141
20	IN	71	118	165
	OUT	94	157	220
25	IN	113	189	264
	OUT	147	245	344
32	IN	181	302	422
	OUT	241	402	563
40	IN	317	528	739
	OUT	377	628	880
50	IN	495	825	1150
	OUT	589	982	1370
63	IN	840	1400	1960
	OUT	936	1560	2184
80	IN	1362	2270	3178
	OUT	1509	2515	3521
100	IN	2145	3575	5005
	OUT	2355	3925	5495

Figure 29. *Theoretical Output for SMC CDQM Compact Cylinder Guide Rod [1]*

Feature	Value
Stroke	300 mm
Piston diameter	50 mm
Piston rod thread	M16x1.5
Cushioning	Pneumatic cushioning, adjustable at both ends
Mounting position	optional
Conforms to standard	ISO 15552
Piston-rod end	Male thread
Design	Piston Piston rod Profile barrel
Position detection	Via proximity switch
Symbol	00991235
Variants	Piston rod at one end
Operating pressure	0.04 MPa ... 1.2 MPa
Operating pressure	0.4 bar ... 12 bar
Mode of operation	Double-acting
Operating medium	Compressed air to ISO 8573-1:2010 [7:4:4]
Note on operating and pilot medium	Lubricated operation possible (in which case lubricated operation will always be required)
Corrosion resistance class CRC	2 - Moderate corrosion stress
LABS (PWIS) conformity	VDMA24364-B1/B2-L
Cleanroom class	Class 6 according to ISO 14644-1
Ambient temperature	-20 °C ... 80 °C
Impact energy in end positions	1 J
Cushioning length	22 mm
Theoretical force at 0.6 MPa (6 bar, 87 psi), return stroke	990 N
Theoretical force at 0.6 MPa (6 bar, 87 psi), advance stroke	1178 N
Moving mass for 0 mm stroke	365 g
Additional moving mass per 10 mm stroke	25 g
Basic weight for 0 mm stroke	1190 g
Additional weight per 10 mm stroke	56 g
Type of mounting	Either: Via female thread With accessories
Pneumatic connection	G1/4
Note on materials	RoHS-compliant

Figure 30. *Festo DSBC-50-300-PPVA-A-N3 [2]*

Similarly, for the upper carrier retrieval system it was important to determine the maximum number of carriers that could be placed in the feed. In this scenario the main concern was failure in steel components used to separate out the carriers. These components will undergo repeated loading during the lifespan of the machine. If the material yields the machine would have to be shut down to be repaired. Halting production. Hence, it is crucial that these components could withstand the load of the carriers. The main components that needed to be analysed where the steel L plates, and the frame which supports the pneumatic actuators. These components would be subjected to much more complicated load cases. Hence, hand calculations and models were used along with finite element analysis (FEA) to increase accuracy.

The steel L plate was modelled as a simply supported 3-D beam with a single load acting at its centre. Where the load had magnitude equal to the weight of 20 carriers. This model is shown in the load case in Figure 31 below. The FEA analysis is shown below in Figure 32, with a scale for the Von Mises stress on the right. Note, the deflection of the components has been exaggerated.

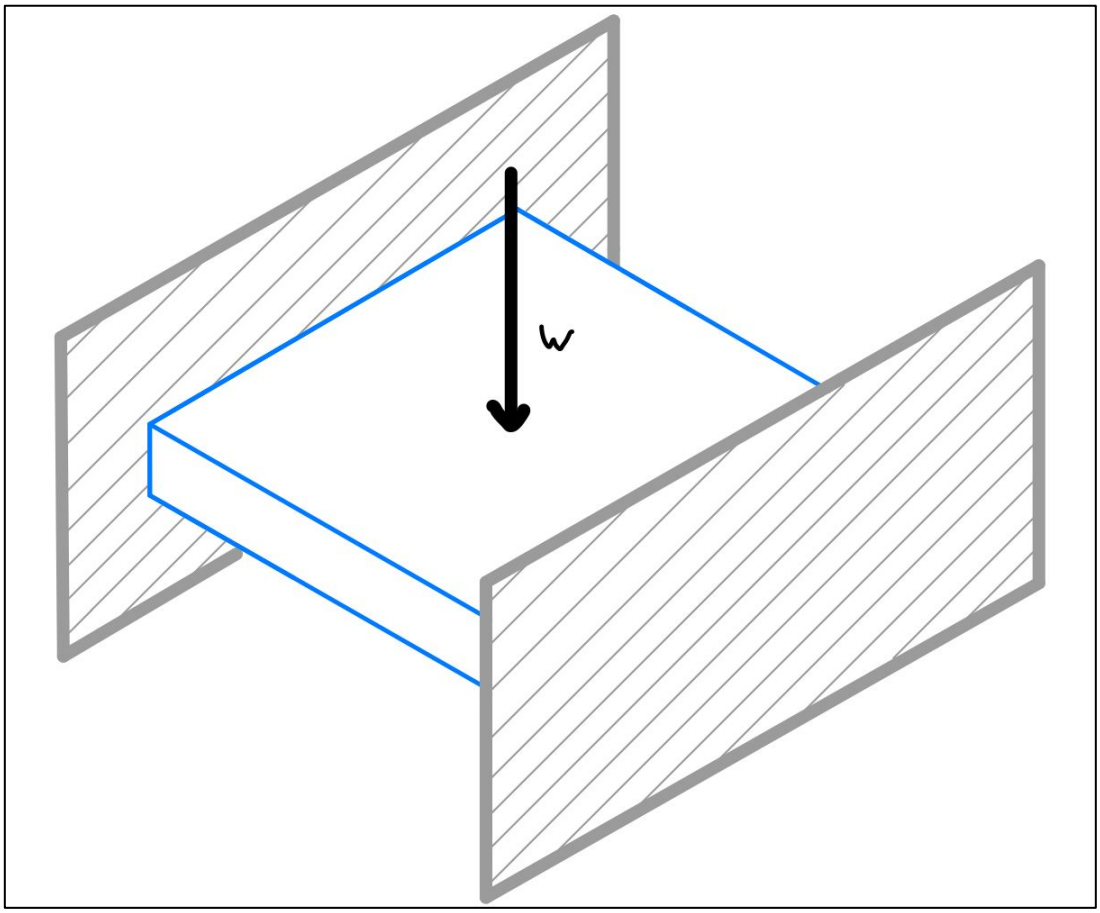


***Figure 31**. Load case for steel L plate*

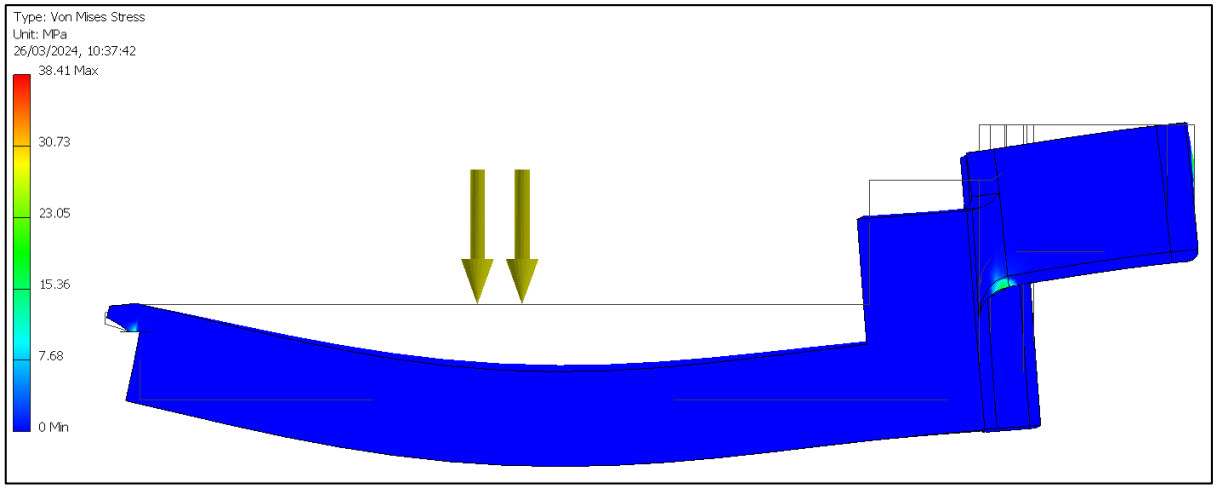


***Figure 32**. Finite Element Analysis on Steel L Plate*

It was determined that the steel clamp met the requirements, the calculations yielded a deflection of 17.96 micrometres. This is supported by the FEA, which shows a low maximum Von Mises stress, and little bending. Meaning steel is a sufficiently strong material to prevent yielding in the component.

For the supporting framework only half the weight of 20 carriers needed to be considered, as the total weight was evenly distributed between two identical frameworks. The FEA was used to determine which beams may be at risk of bending or yielding. Then hand calculations were

used to determine the deflection. The FEA for the support frame is shown below in Figure 33. Note for complicated structures such as the one being tested results are not reliable. This is why calculations were used in conjunction.

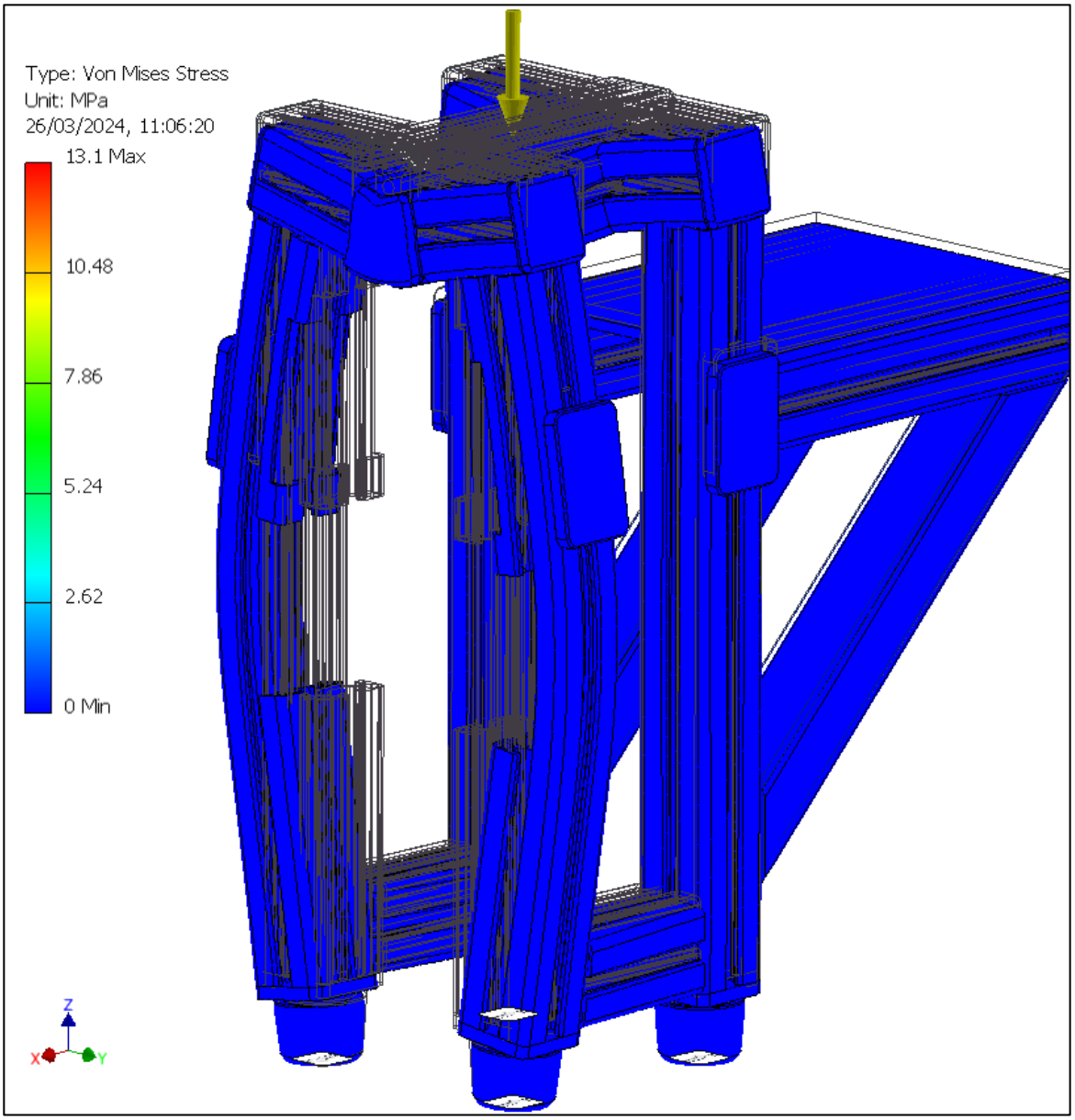


Figure 33. *Finite Element Analysis on Pneumatic Actuator Support Frame*

Figure 33 shows that the top horizontal beams where at greatest risk of bending. All three beams at the top had near identical dimensions. For this reason, bending calculations only had to be done on one. These calculations are shown in Appendix C. Where it was modelled as a simply supported 3-D beam with a single load acting at its centre, this model is shown below in Figure 34. When using the total weight of the carrier (double the magnitude the support frame would experience) the calculations yielded a maximum deflection of 6.857 micrometres. Meaning there is little risk of the support structure bending during operation.

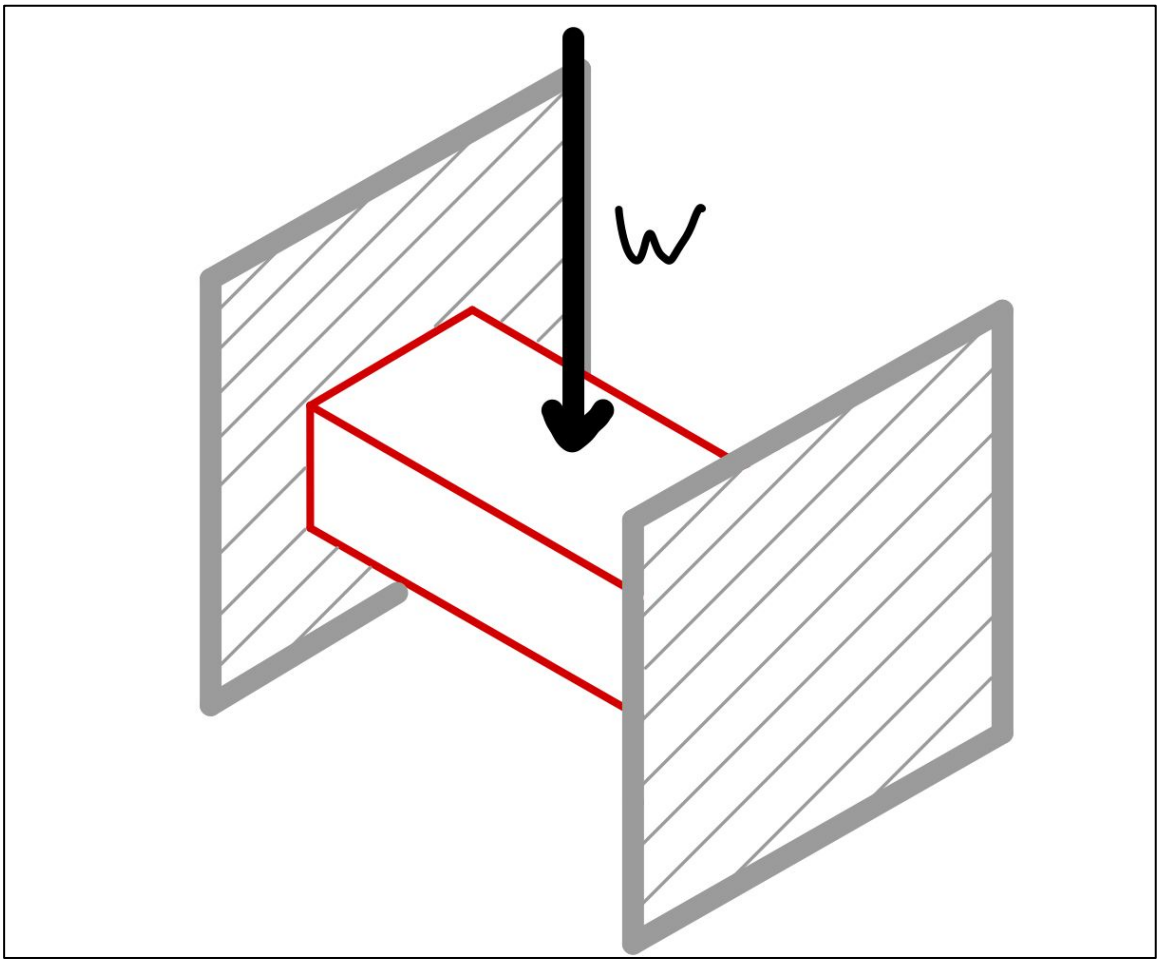


Figure 34. *Finite Element Analysis on Pneumatic Actuator Support Frame*

Following the insertion of the control rods into the lower carrier, the half-assembled module is then transported under the upper carrier retrieval mechanism. This is achieved using a pneumatic actuator which pushes the half-assembled module. This process needs to be precise so that the slots in the upper carrier are aligned with the control rods. This stage is crucial for the assembly process. Any misalignment between the control rods and upper carrier could result in damage to the control rods or carrier and prevent completion of the module. For this reason, it is important to evaluate the friction between the half-assembled module and the ground to reduce the effect of disturbances.

A 2-D force diagram was used to determine the friction on the half-assembled module. The load case is shown below in Figure 35. The calculations are shown in Appendix C. Then, by comparing the maximum output force of the Festo DSBC-50-300-PPVA-A-N3 that was selected (P) to the friction (F) it was determined that the friction would cause minimal disturbance and was not enough to influence the operation of the mechanism.

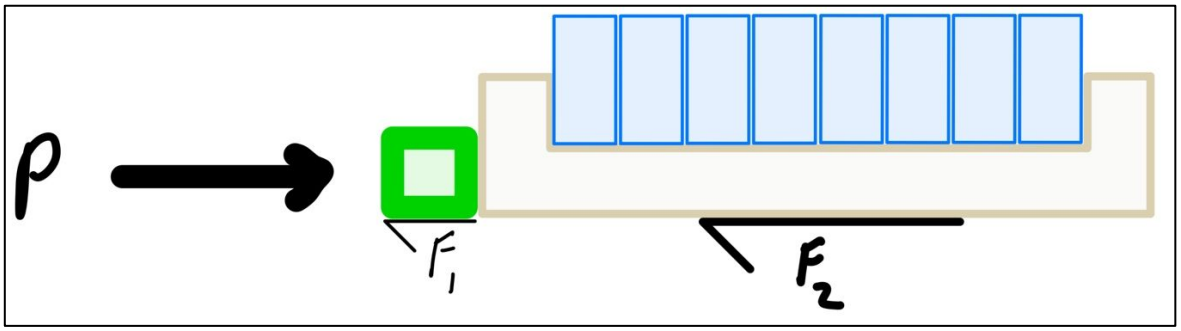


***Figure 35**. Load Case for Transfer Stage Pneumatic Actuator*

Before the rod insertion, control rods accumulate at the end of the buffer. It was important to determine the total number of control rods the buffer could handle. If the motor powering the magnetic conveyor belt was overloaded, the entire buffer would have to be replaced. It was determined that a maximum of 60 rows of control rods could be used before the motor failed. Which is more than the buffer can physically hold.

Production Calculations

The process flow diagram below in Figure 36 shows the main stages of the machine and their sequence.

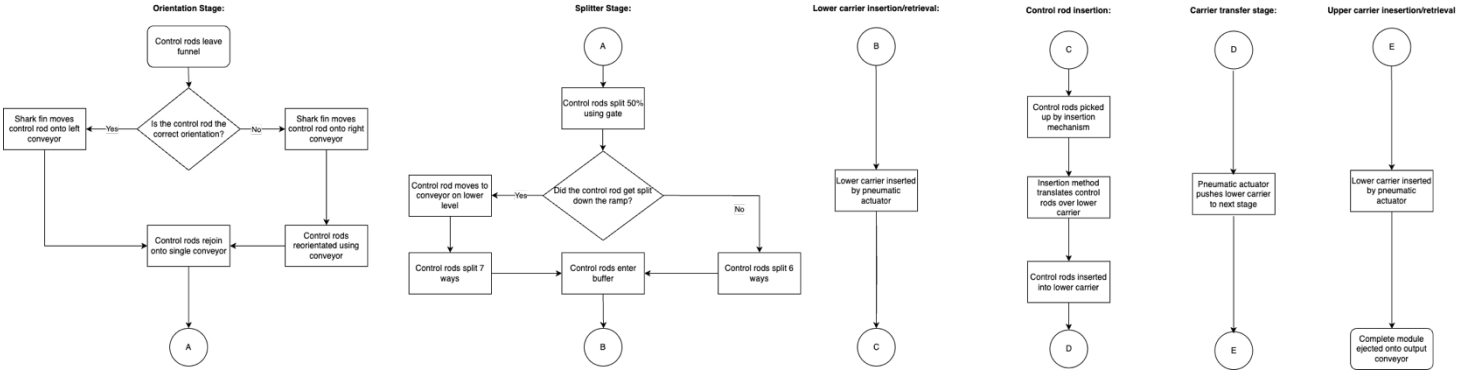


***Figure 36**. Process Flow Diagram of Machine*

The process flow diagram seen above was used as the basis for the timing calculations of each operation within the machine. A simplified route of a control rod was then created and used to find the timings. The route and supporting calculations are shown in Appendix E. Given that the average throughput of the control rods is 1800 rods/s, the following timings are yielded:

- Time taken for a control rod to reach the buffer of 2.45 seconds
- Time for one row of the buffer to be filled of 26.95 seconds
- Time for the lower carrier to be inserted of 0.24 seconds
- Time for rods to be inserted of 4.10 seconds
- Time for carrier to be transferred between stages of 0.021 seconds

The timings could then be used to find the cycle time and production rate. Note, the orientation stage and splitter stage must occur in order. These stages are the limiting factor for the cycle time as the next stages must wait for the buffer to fill before beginning. However, while the buffer fills the lower carrier can be inserted, cutting down the cycle time. Given that the insertion time of the control rods is much lower than the time for one row of the buffer to be filled, the insertion time for the control rods can be neglected. This is because by the time the first row has been inserted into the carrier the second row has not finished entering the buffer. Furthermore, the upper carrier time can be neglected. This yields the following:

- A Cycle time of 136.4 seconds
- A Production rate of 26.4 modules per hour

Which is significantly greater than what is required by the client. However, this allows the annual production quota of the client to be met even if there is a fault with the machine, or maintenance takes an extended period.

The time between carrier resupplies was then calculated. This was found to be:

- A maximum time to resupply of 45.5 minutes

The machine is heavily reliant on a continuous supply of control rods. Meaning the machine is easily influenced by any variation in the control rod throughput. This is because the machine relies on a minimum of 13 control rods to fill a buffer before inserting them into the lower carrier. This is what limits the production rate down the most. Hence, calculations to determine the balancing loss of the machine were done. This yielded:

- Balancing loss of 99.98%

Which is to be expected, as majority of the time the machine is waiting idle for the buffer to be filled. However, the buffer has the benefit of being able to manage high control rod influx without overflow.

The timing calculations made the following assumptions. For the conveyor timing the following assumptions were made:

- Conveyor belt operates at constant speed of 2m/s
- No slip between control rods and belt
- Curved conveyor belt distance is small compared to straight aways
- All control rods travel the same distance

For the control rod insertion:

- Linear unit operates at a constant speed of 5m/s

- Actuator operates at a constant speed of 50mm/s

For the lower carrier insertion/transfer:

- Constant acceleration

For the upper carrier insertion:

- Time to insert upper carrier is negligible

Design Iterations

Input Conveyer and Funnel:

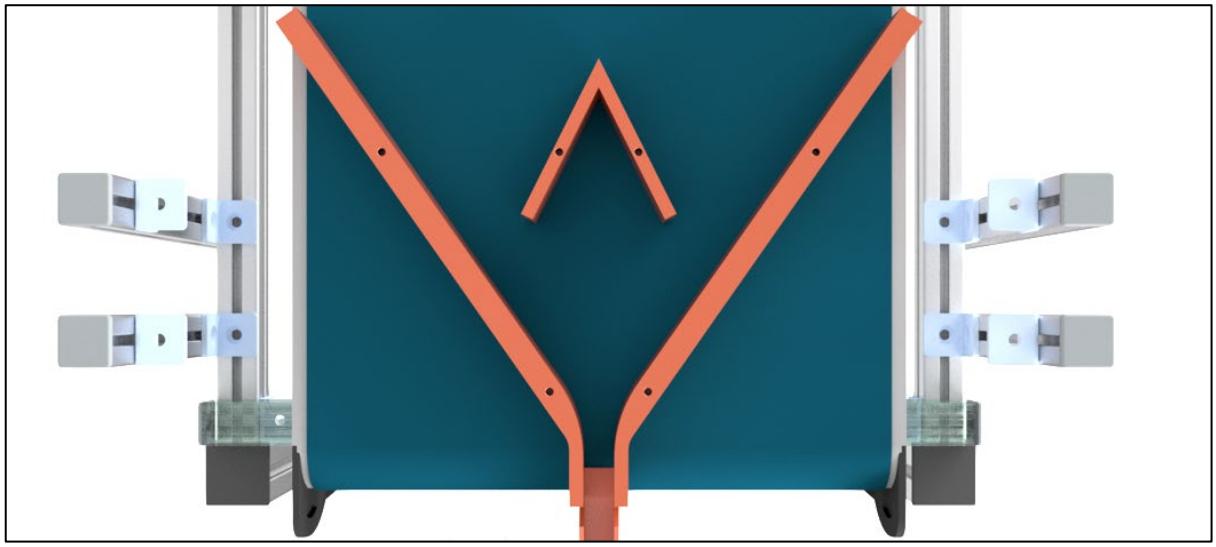


Figure 37. *Rod Funnel (some beams were hidden to aid visibility)*

The funnel over the input conveyor was modified by giving the outer funnel a shallower angle and adding a large radius leading into the narrowest part of the funnel (Figure 37); this results in a more gradual change in angle to the rod as it is funnelled reducing bottle necking and jamming. The use of the central element helps to prevent multiple rods entering the exit passage at the same time and stops rods entering at an angle perpendicular to the passage as these could both jam the system.

Rod Orientation:

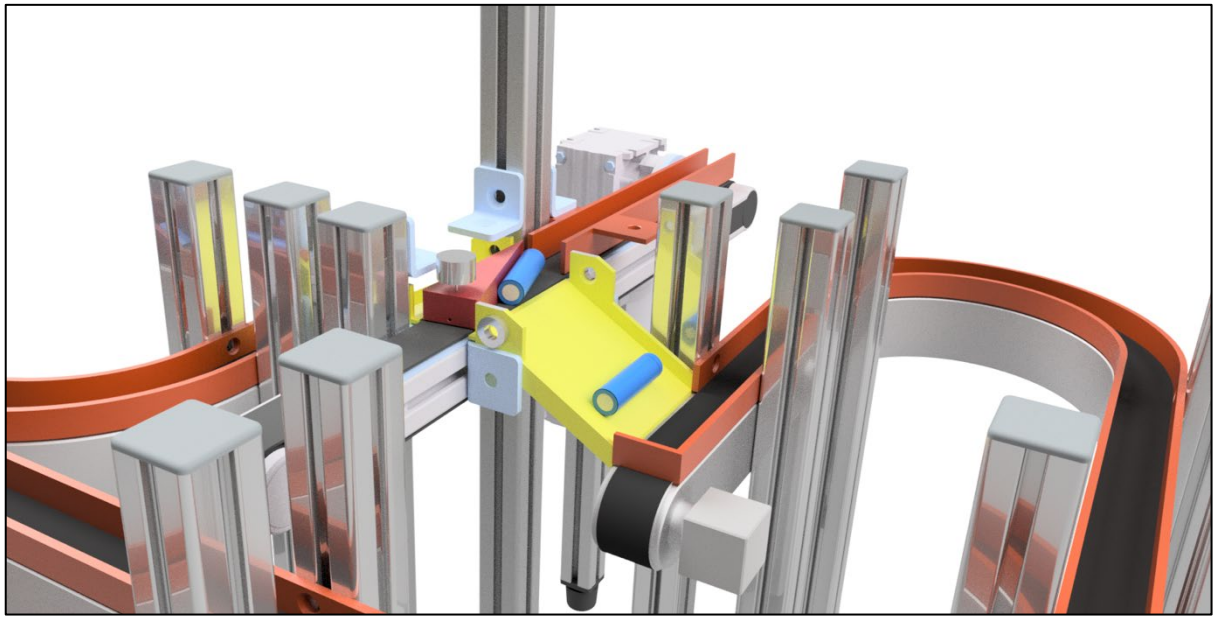


Figure 3810. *Rod Scanning and initial splitter (covers were hidden to ease visibility)*

Guide rails were added to all the conveyors to prevent rods from falling onto the ground (Figure 38). In addition to this the module was made larger to accommodate for the custom Monk Flexible Conveyor Belts with a curve radius of 150 mm. It was decided that 40 x 40mm profile beams would be used to support the conveyors and guides due to their versatility of bracket mounting options.

Rod Splitting:

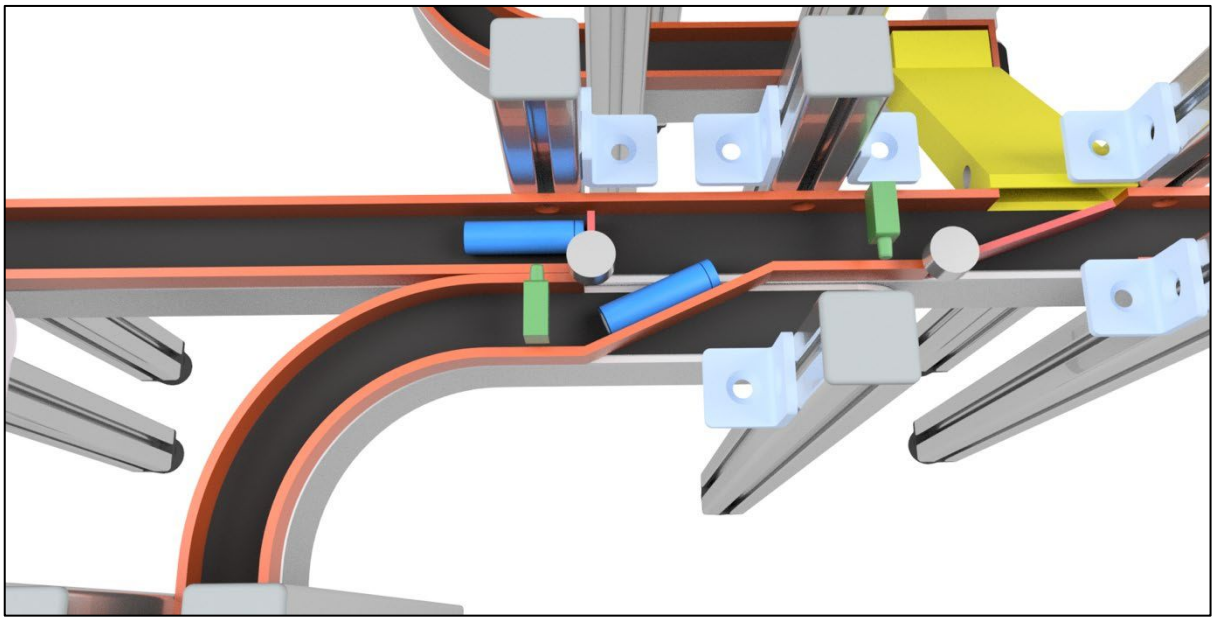


Figure 39. *Rod re-joining and subsequent splitting (some beams were hidden to ease visibility)*

After the rods are oriented in the same direction, they are re-joined onto the same conveyor belt (Figure 39). At this point the actuator from the initial design concept (Figure 7) was replaced with a fin to block the rods in one of the channels while the other can flow freely. This allowed for a more reliable continuous flow of rods, where it is not necessary to wait while an actuator retracts. Next the rods are split 6-7 to the upper and lower conveyors, here a fin also guides the rods onto a ramp if they are to be sent to the lower conveyor.

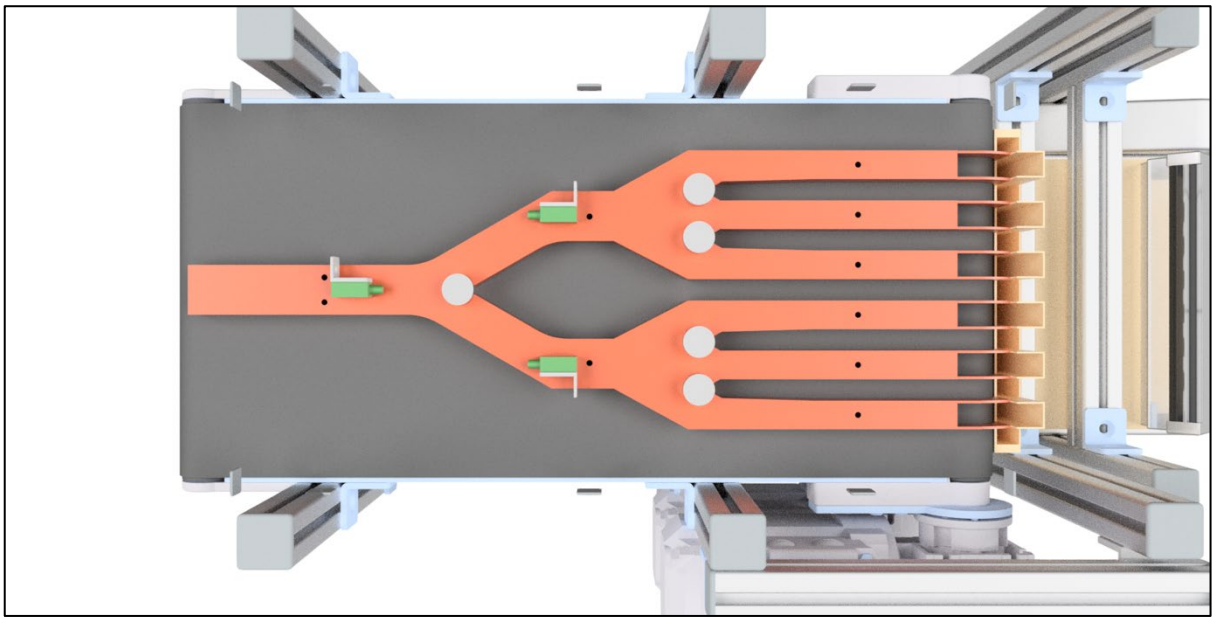
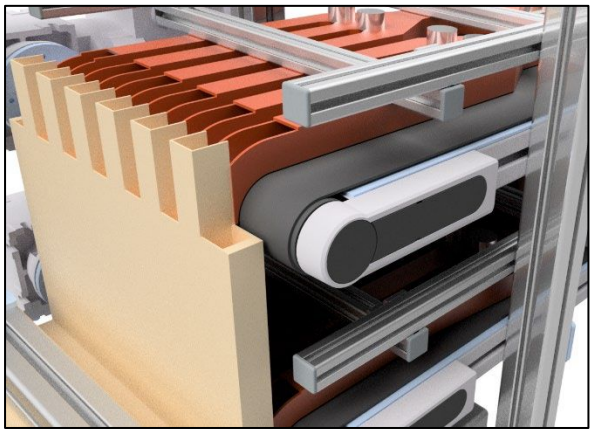


Figure 40. *Rod 6-way splitter (some beams were hidden to ease visibility)*

The splitters were changed through many iterations to create a smooth passage for the rods where they would not be blocked. To prevent jamming of rods inside the channels as they are split, the channels were shaped with curved edges in specific places to allow a rod to easily change direction and move to the correct passage. As a result of this design the part is to be custom machined from

plastic. The splitters were also changed to be covered over the top; this allowed motors to be mounted onto them which control the guiding fins, as well as beams to hold them in place.

Rod Buffer:



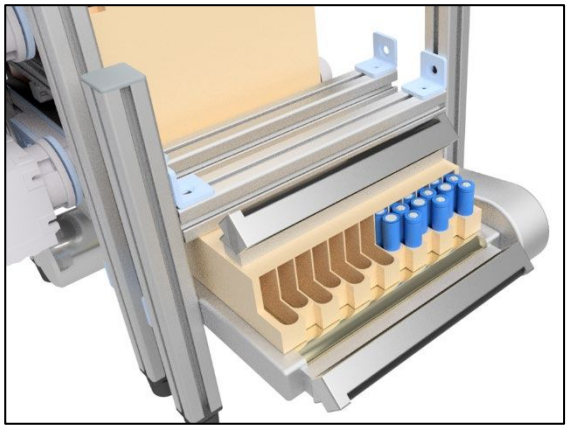


Figure 41 & 42. *Splitter end and Buffer end*

At the end of the splitters the walls curve down, shown in Figure 41, following the profile of the conveyor belt into the Buffer Shoots where rod fall onto a magnetic conveyor; this change was made to rotate the rods gradually to a vertical orientation stopping them from jamming in the buffer. Once in the buffer the addition of a magnetic conveyor prevents the rods from falling over as they move but the conveyer can still slide beneath the rods once they accumulate at the end. The profile at the end of the buffer shown in Figure 42 has the same spacing as the Carriers and the rod Presser so the rods can be picked up with precision.

Rod Insertion System:

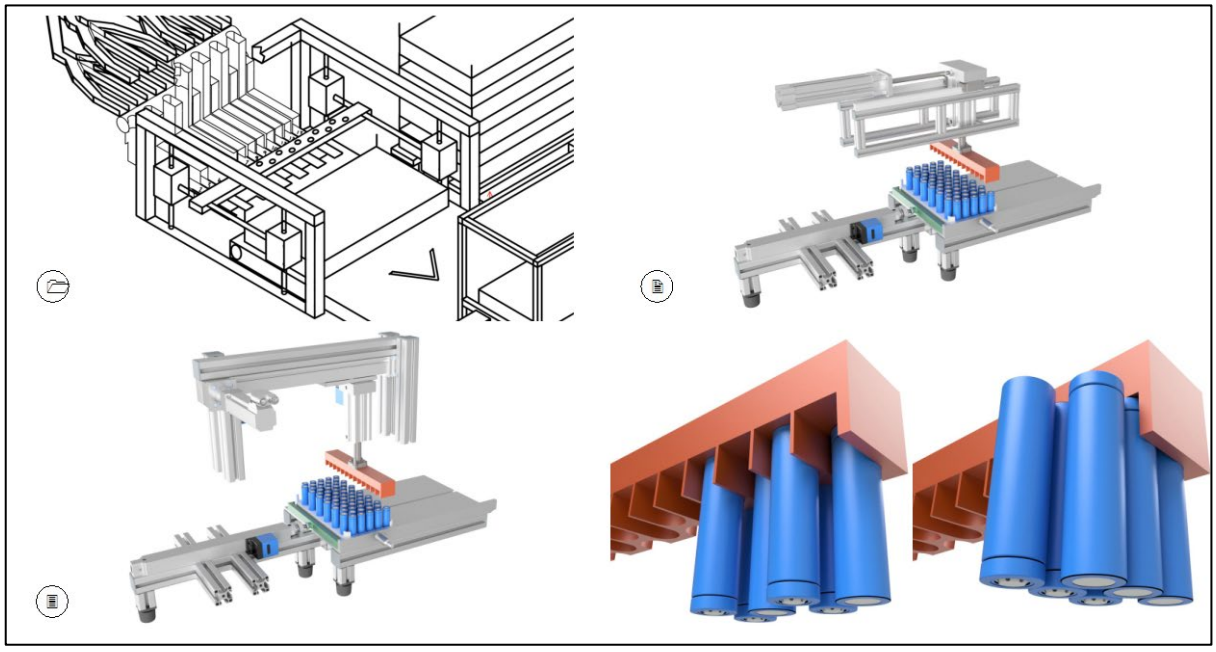


Figure 43. *Rod presser design evolution*

The Rod Insertion System saw the most changes out of all the subsystems. Due to the availability of off-the-shelf components, the original layout for the Rod Insertion Mechanism was changed. Instead of the horizontal actuator being pushed down as the Rods are inserted, the vertical actuator would be

moved horizontally, and the horizontal actuator would remain stationary. This led to the second iteration of the system. The second iteration came when it was decided to switch from an actuator to a linear slide unit for the horizontal motion of the Presser. This was done due to a lack of space, as the actuator would have interfered with the Rod Buffer. The linear slide unit also eliminated the need for the load carriages and rails further reducing cost as the slide unit can carry the weight of the Presser System. The Presser itself did not see much change; its shape was refined when the CAD model was built. Its geometry allows it to pick rods up without interfering with the other rods in the buffer, and to press them into the carrier without interfering with rods already in place.

Bottom Carrier Input System:

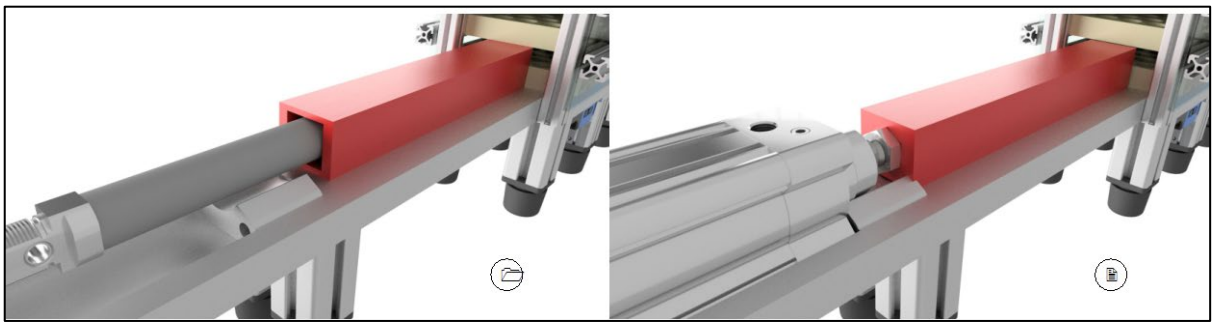


Figure 44. *Bottom Carrier Piston design evolution*

The design of the Bottom Carrier input system remained mostly unchanged from the initial concept, the sizes of some of the custom components were changed to accommodate the use of off-the-shelf actuators and extruded frames. In the original concept, the piston (depicted in red) would fit over the actuator pushing it when retracted. This required the piston to be hollow, which would compromise its structural integrity when loaded with the weight of the Lower Carrier stack. Furthermore, after selecting a suitable actuator, it was too large for the piston to move over it without changing the height of the piston. This would stop the piston from fitting under the remaining Carrier stack after pushing a carrier into place. The piston was changed to be a solid cuboid with the height of a carrier, so the stack of carriers does not fall when the bottom most carrier is extracted. This redesign moved the actuator further out, increasing the overall width of the machine, but the new footprint was still within the specified floorplan area. This also decreased the manufacturing costs of the custom component.

Carrier Movement System:

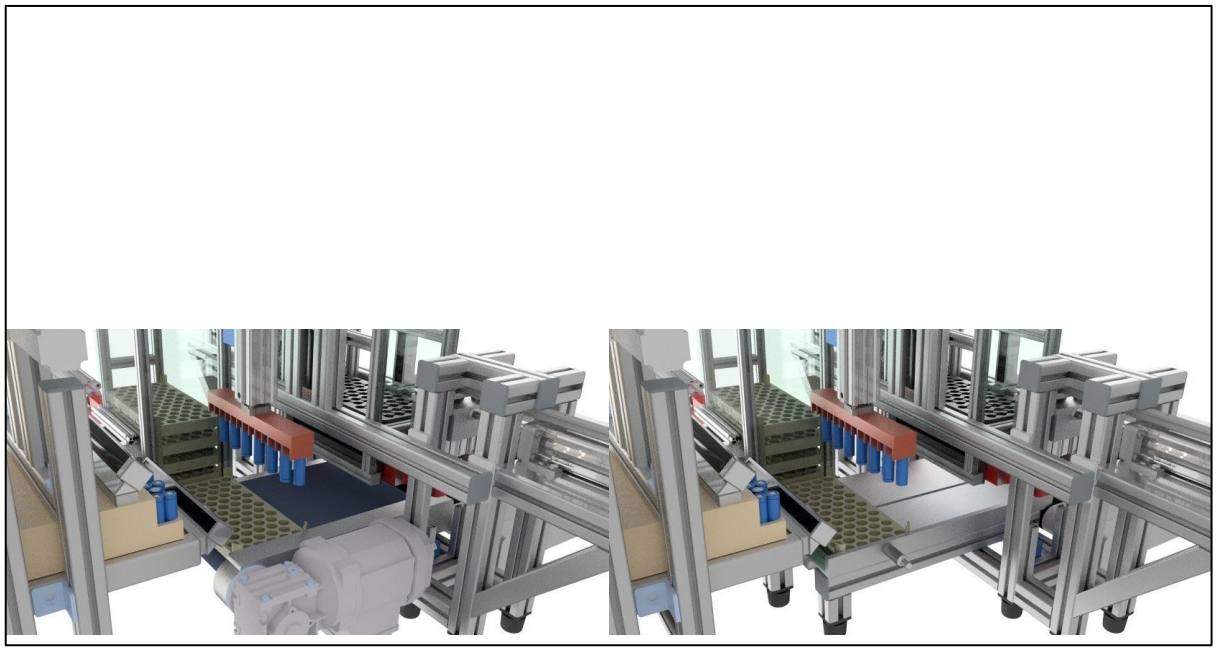


Figure 45. *Carrier Movement System design evolution*

This system (Figure 45) was changed from a timing conveyor belt to an actuator, that pushes the filled Bottom Carrier to where it is mated with the Top Carrier. This change was made to increase the reliability of the system, as the extension of an actuator provides more precise movement than a conveyor belt, and to decrease cost, as an actuator and a simple aluminium base are less expensive than a precision conveyor belt system. This also allowed for a less powerful and expensive actuator to be used for the extraction of the Bottom Carrier from its stack as the Carrier would no longer have to overcome the friction from the belt when sliding out.

Upper Carrier Input System:

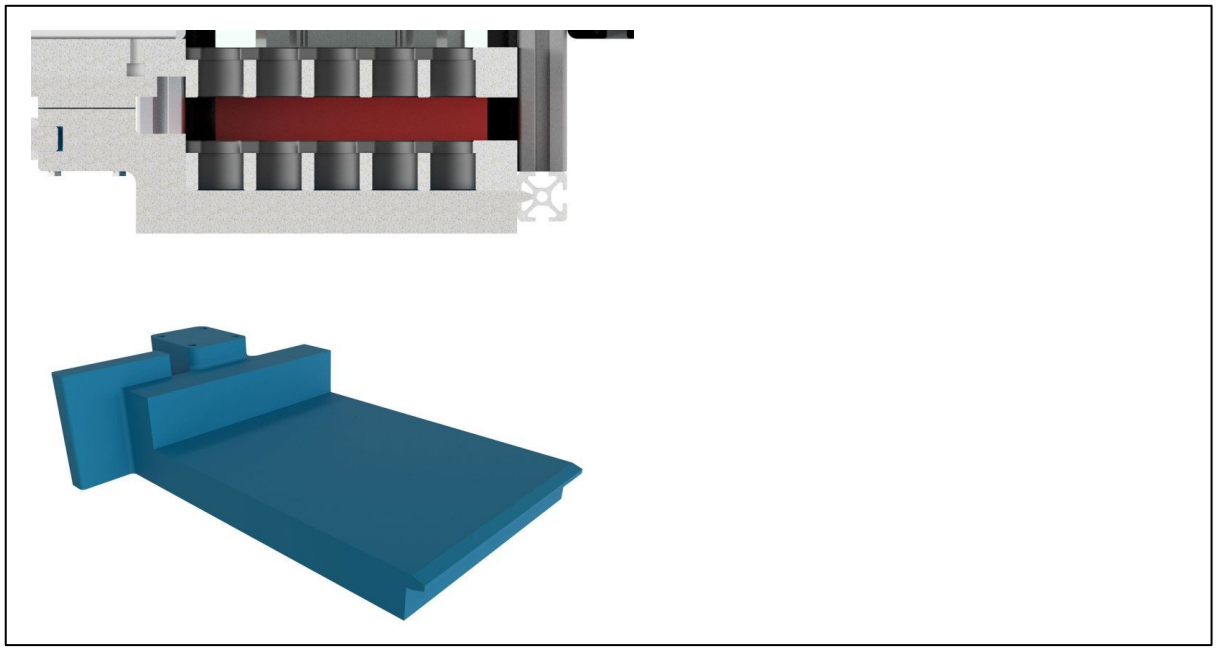


Figure 46. *L plate geometry and fit with extruded frame and Clamp geometry*

The Upper Carrier Input system did not evolve much from the initial concept, most changes were made to the geometry of the L plate and Clamps. The lower plate of the Clamps was extended to increase the surface area that the carrier rests on. Holes matching the size and pattern in the Lower Carrier were cut into the extended plate so that it would not interfere with Rods in the Bottom Carrier when the Upper Carrier is mated. A small extrusion was added to the end of the L plate to fit into the cut-out in the selected aluminium extrusion; this was done so that the plate would have two contact points supporting the weight of the Upper Carrier stack when extended. A small plate was also extruded from the L plate to allow the ultrasonic waves from the sensor to bounce back, to measure the distance.

Final Design Solution Specification Table

FDSST
The machine runs 24 hours, 7 days a week
The lifetime of the machine is of 10 years
The overall dimensions of the machine are, length: 4997 mm, width: 1859 mm, height: 944 mm
The total area of the machine is 9.289 m^2
The total cost of the machine is £20,746 (excluding labour costs)
The module assembly cycle is of 136.4 seconds
Operating pressure: 6 bar
All parts of the machine are accessible to an operator at standing level
Machine presents low risk to operator during operation
Components are easy to access and disassemble and are standard

Figure 47. *Final Design Solution Specification Table*

Reflections and Conclusion

To conclude, the design achieves and surpasses many of the requirements set at the beginning of the project. The machine has a cycle time of 136.4 seconds. Most of the machine's operations are monitored by a range of sensors that can alert the technicians should any problem arise. The thorough and methodical design process makes it simple to review the thought process behind the design and provides a clear justification for every engineering choice. This aided in identifying any design flaws or errors that were later corrected with further iterations of the machine. Calculations were made for the critical systems of the machine and components selection was done with the aid of calculation to determine the requirements the off-the-shelf component had to meet.

A lot of aspects in this project went well, the team worked efficiently, as reflected in the Gannt chart, and all tasks were completed in full and on time. The concept selection was objective, and the best solutions were chosen for the various mechanisms of the machine which led to a functional and relatively simple machine design final concept. The reliance on custom components was minimised and they were designed to function within the specified design parameters as seen from the calculations. However, the machine was over the budget, going over by approximately 5,000-pounds (check Appendix F for Cost Calculations). The main contributor to the cost was the conveyor belts and actuators in the design which were more expensive than anticipated. To ensure the next project does not go over budget, a very rough cost estimation could be done once the final concept is chosen; the average cost for the main components of the system could be found to give an idea of the full cost of the machine. In this case, the average cost of a conveyor belt and of a pneumatic actuator could have been found and then multiplied by the amount of each item to give a cost estimation, the design would have then been changed accordingly to remain within budget. But overall, the project was completed successfully, efficiently and within the required time frame with little issues arising during the design process.

References

1. SMC CORPORATION, "Compact Cylinder/Guide Rod Type," CAT.EUS20-179 B-UK
2. Festo, "ISO Cylinder DSBC-50-300-PPVA-N3," 2098975, Mar. 2024

Appendix

Appendix A: Gantt Chart and PDS

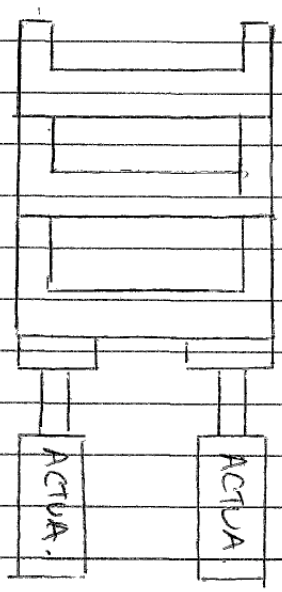
	Week	W1	W2	W3	W4	W5	W6	W7	W8
Activity									
PDS									
Machine requirements from brief									
Good machine design consideration									
Final PDS overview									
Morphological functions									
Complete function sketches									
Choose functions									
Concept sketches									
Final concept drawing									
UML diagram									
Method of Operation									
CAD models									
Module layout CAD									
Design modules									
Import selected components									
Component selection									
Select actuators									
Select frame profiles									
Select conveyer belts and rollers									
Select sensors									
FMECA									
FTA									
Report									
Design Freeze Report									
Technical Report									

Figure 48. *Gantt Chart of Project Evolution*

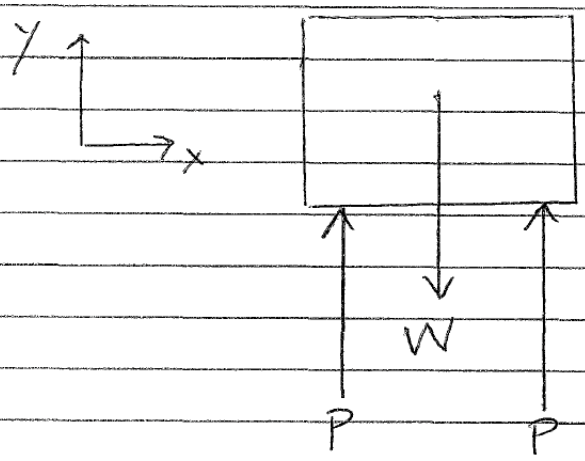
#	Category	N / W	Wt	Requirement	Source	Date
1	Overall Function	N		The machine must be autonomous	Client	06/02/2024
2		N		The machine must operate in a 10 m^2 footprint	Client	06/02/2024
3		W	M	Minimise energy consumption	Team	06/02/2024
4		N		Target cost is £15,000	Client	06/02/2024
5		N		Time taken to assemble one control rod module should not exceed 180 seconds	Client	06/02/2024
6		N		The machine must operate for at least 10 years, with three 8 hour shifts per day	Client	06/02/2024
7		N		The machine must operate at a minimum of 13°C	Client	06/02/2024
8		N		The machine must operate between 40% and 70% humidity	Client	06/02/2024
9		W	L	Rod input must be close to completed module output	Client	06/02/2024
10		W	M	Machine sub-sections must be modular	Team	12/03/2024
11		W	M	Individual machine components must be easy to disassemble and to replace or reuse after end of life	Client	06/02/2024
12	Safety, Maintanence and Human Factors	N		The machine must operate with a maximum noise level of 85 dB	Client	06/02/2024
13		N		The machine must be accessible by humans to service it twice per year	Client	06/02/2024
14		N		The machine must be safe for an inspector and mechanic to work on it when turned off	Team	12/03/2024
15		W	H	All parts of the machine should be accessible to the operator at standing level	Team	08/02/2024
16		W	H	Standard fasteners must be used to ease assembly and disassembly	Client	06/02/2024
17		W	M	The machine must use standard components where suitable to minimise costs and for ease of maintenance.	Client	06/02/2024
18		W	H	The machine must halt when a failure is detected	Team	12/03/2024
19		W	H	The operator should not need to move around a lot while operating the machine	Client	12/03/2024
20	Funnelling Rods	N		The machine must accept input from a 500mm wide belt	Client	06/02/2024
21		N		The machine must be able to handle an average of 1800 rods per hour	Client	06/02/2024
22		N		The machine must be able to arrange rods from a disoriented manner into an organised input	Team	06/02/2024
23		N		The machine must not get jammed when multiple rods are inputted simultaneously	Team	11/02/2024
24		N		Rods must be prevented from falling from the conveyor belt	Team	06/02/2024
25	Sorting (scanning) and re-orienting (flipping) rods	N		The rods must be positioned 300mm from the lense of the camera and be within the camera frame	Optronis	06/02/2024
26		N		The rods orientation must be scanned	Team	06/02/2024
27		N		Rod orientation must be able to be changed by 180° to allow for the rod insertion pattern	Team	06/02/2024
28	Lower carrier input	N		Lower carriers must be retrieved from a buffer (container / magazine) of carriers	Client	06/02/2024
29		N		Retreived carrier must be ready to receive rods	Team	06/02/2024
30	Lower carrier rod placement	N		The rods must be placed into the carrier in 13 rows of 5 in alternating orientations	Client	06/02/2024
31		N		Rods must be positioned with spacing in accordance with figure 5	Team	06/02/2024
32		N		Minimum force of 50N/rod to secured rods into the lower carrier	Client	06/02/2024
33	Upper carrier input	N		The machine must retrieve an upper carrier from a buffer of upper carriers	Client	06/02/2024
34		W	L	Upper carrier input must be close to lower carrier input	Team	06/02/2024
35		N		The upper must be alligned with the lower carriers before placement	Team	06/02/2024
36	Upper carrier mate with lower carrier	N		The upper carrier must be guided into place with the lower carrier to allow for insertion of rods into the upper carrier	Team	06/02/2024
37		N		Minimum force of 50 N/rod force to secure upper carrier	Client	06/02/2024
38	Final product output	N		The machine must eject the completed rod module using a conveyor of width 500mm	Client	06/02/2024
39	Rods	N		Rods must not be contaminated	Client	06/02/2024
40		N		Rods must be handled with care	Client	06/02/2024
41		N		There must be a ±0.5 degrees tolerance for inserting the rod in carrier	Team	06/02/2024

Figure 49. *Product Design Specification (PDS)*

Appendix B: Bottom carrier retrieval calculations



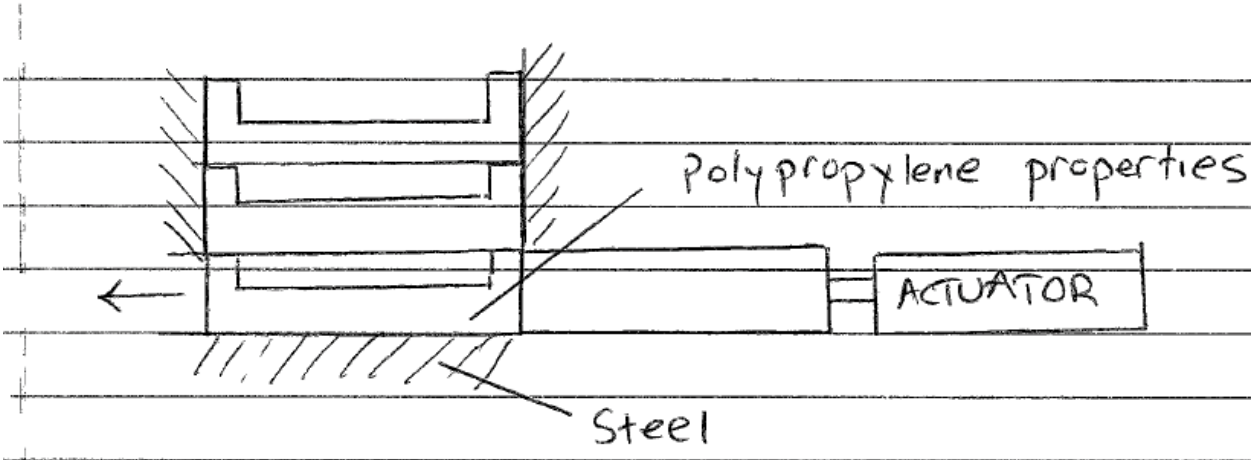
F. B. D



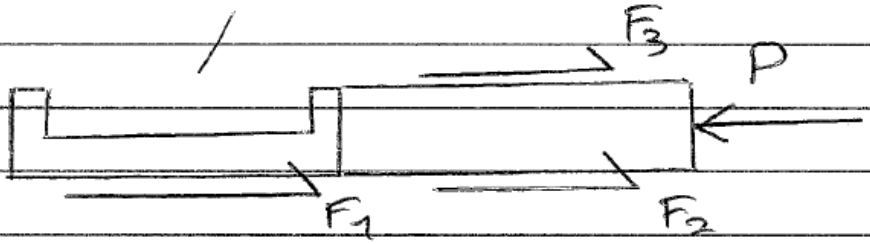
W for 20 carriers = 67.8852N
W for 100 carriers = 339.426N

Min force for $P = \frac{W}{2}$

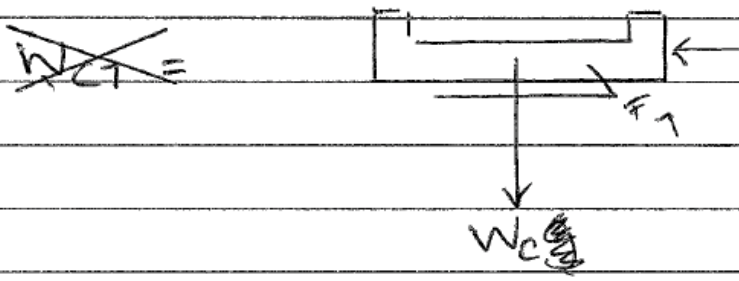
$P = 33.94N$ or $169.7N$
(20 carriers) (100 carriers)



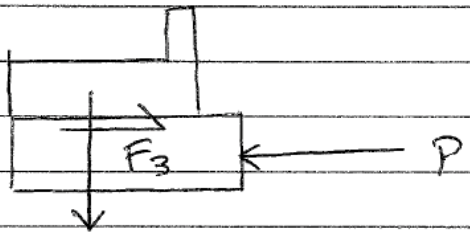
F.B.D



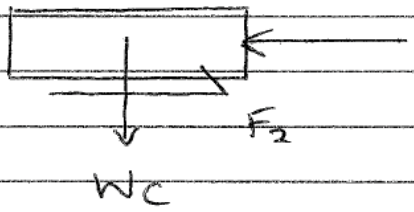
Friction on upper surface of carrier considered to be negligible compared to friction on upper surface of pusher.



W_c = weight of carriers in feed (20 oz



W_{c1} = weight of 1 carrier
W_c = weight of carriers in feed (20 or 100)



Coefficient of friction between polypropylene and steel

$$\mu_p = 0.42$$

Coefficient of friction between steel and steel

$$\mu = \mu_s = \mu_p = 0.42$$

$W_c = 339.426\,N$
$W_{c1} = 336.032\,N$

For 100 carriers

$F_1 = \mu R_1$ $R_1 = W_c$ for 100 carriers

$F_1 = 0.42(339.426)$

$F_1 = 142.6N$

$F_2 = \mu R_2$ $R_2 = W_c$ for 100 carriers

$F_2 = 0.42(339.426)$

$F_2 = 142.6N$

$F_3 = \mu R_3$ $R_3 = W_c - W_{cn}$ for 100 carriers

$F_3 = 0.42(336.032)$

$F_3 = 141.1N$

$\Sigma F = 142.6 + 142.6 + 141.1$

$= 426.3\ N$

For 20 carriers

$W_C = 67.8852\,N$

$W_C - W_{C1} = 64.491\,N$

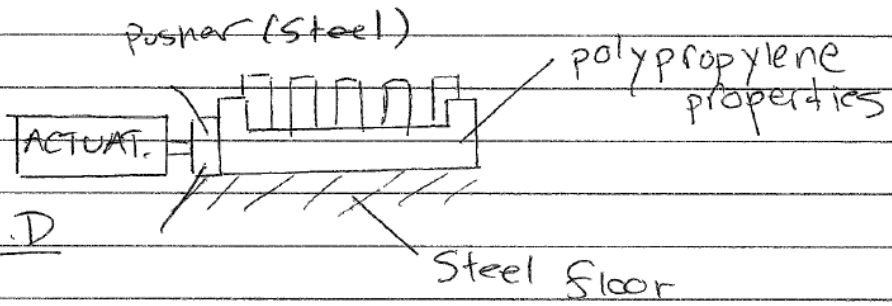
$F_1 = F_2 = 28.5718$

$F_3 = 27.0862\,N$

$\Sigma F = 2F_1 + F_3$

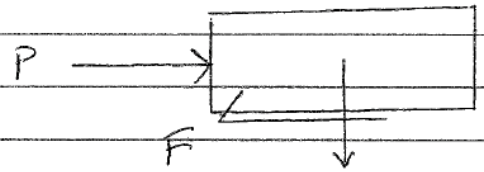
$F = 84.11\,N$

Appendix C: Lower carrier transfer calculations



F.B.D

Steel floor



Coefficient of friction between polypropylene and steel:

$\mu = 0.420$

NB// This is the same for coated steel on coated steel

$W_c = 0.346\,kg$ → ~~weight~~ mass of carrier

$W_p = 0.752\,kg$ → mass of pusher

$W_R = 65(0.135) = 8.775\,kg$ → mass of 65 control rods

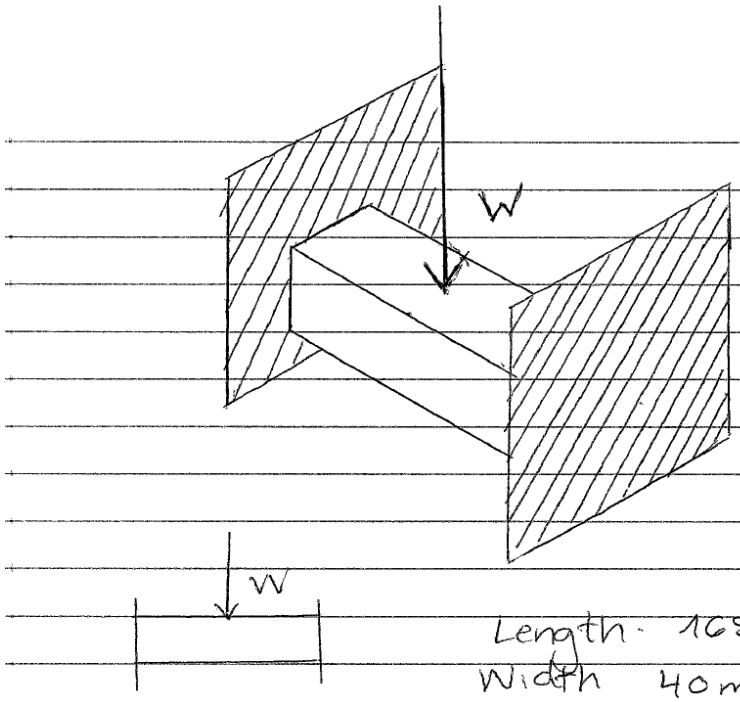
$F = \mu R$ } $R = (W_c + W_p + W_R)g$

$F = 0.420(0.346 + 0.752 + 8.775)(9.81)$

$F = 38.2N$

∴ Friction provides a negligible disturbance to the pneumatic actuator

Appendix D: Upper carrier transfer calculations



Length: 165mm
Width 40mm
Height 40mm

Deflection of beam when simply supported: $\delta = \frac{P \cdot L^3}{48 \cdot E \cdot I}$

$P = w = 67.8852N$ → weight of 20 carriers

$E = \frac{\cancel{215}}{69}$ GPa → Young's Modulus of Aluminium

I = moment of inertia

L = Length

$$I = \frac{b \cdot h^3}{12} \quad \begin{cases} b = 40 \\ h = 40 \end{cases}$$

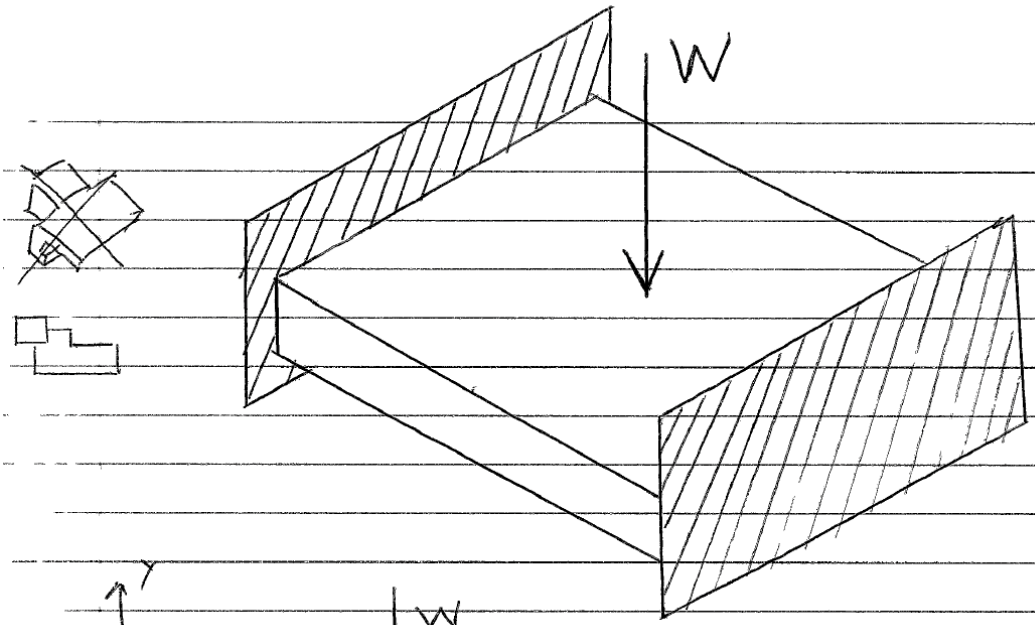
$$I = 21333.33 \text{ mm}^4$$

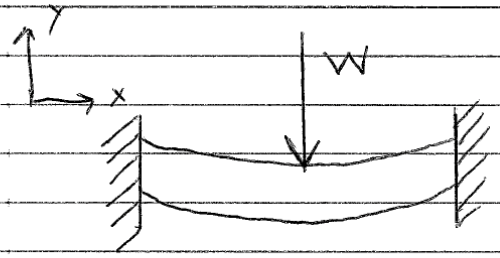
$$\delta = \frac{67.8852(165)^3}{48 \times 69 \times 10^9 \times 21333.33}$$

$$\delta = 6.857 \times 10^{-6}$$

or

6.857 micrometers





Length 134.950mm
Width 110mm
Height: 12.629mm

Deflection of beam when simply supported.

$$\delta = \frac{P \cdot L^3}{48 \cdot E \cdot I}$$

$P = W = 67.8852N$ → weight of 20 carriers
$E = 215$ GPa → Young's Modulus of steel
I = moment of inertia
L = length

$$I = \frac{b \cdot h^3}{12} \quad \left\{ \begin{array}{l} b = 110mm \\ h = 12.629mm \end{array} \right.$$

$$\therefore I = 190158.8 mm^4$$

$$\delta = \frac{67.8852\,(134.950)^3}{48 \times 275 \times 10^9 \times 190{,}158.8}$$

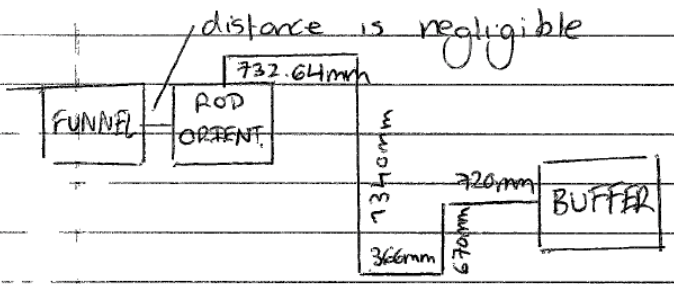
$$\delta = 1.796 \times 10^{-5}\ m$$

or

17 96 micrometers

Appendix E: Timing Calculations

Funnel → Rod Orientation → Splitter (Buffer)



Simplified model of control rod path.

∴ Distance traveled by control rod

= 3.83m

Taking speed of the control to be

= $2.0ms^{-1}$

$$\text{Speed} = \frac{\text{Distance}}{\text{time}}$$ assuming constant speed.

$$\text{time} = \frac{3.83}{2.0}$$

= 1.915 seconds

Control rods also travel via conveyor in buffer. The approximate distance traveled is

= 1.070m

∴ time = 0.535 seconds

∴ The time taken for one rod to reach the splitter/buffer is $t = 2.45$ seconds.

~~Given~~

Time for buffer to fill one row

Given that rods enter the machine with a frequency

$= 0.5$

$0.5 \times 2.45 = 1.225$ balls reaching the buffer every 2.45 seconds

$\frac{13}{1.225} = 10.61$ - It takes 11 'cycles' to fill the row

Meaning the time taken to fill one buffer row is

~~13~~ $11 \times 2.45 = 26.95$ seconds

Time for lower carrier to be inserted

Actuator stroke 300mm
Output force of actuator: 1178N max at 6 bar

Friction for 20 carriers 84.71N
Friction for 100 carriers 426.3N

Maximum time taken for lower carrier to be inserted when friction max.

Σ forces = ma

$1178 - 426.3 = 0.346a$

$a = 2172.5\ ms^{-2}$

Which would damage the carrier and is overkill.

For this reason actuator limited to output force of 430N

Yielding an acceleration of

$10.7\ ms^{-2}$

Assuming constant acceleration

$s = ut + \frac{1}{2}at^2$ $u = 0$, $v = 0$

$300 \times 10^{-3} = \frac{1}{2}(10.7)t^2$

$\therefore t = 0.237$ seconds

Time for rods to be inserted

Linear unit speed $= 5 ms^{-1}$

Linear unit length $= 524 mm$

$\text{time} = \frac{524 \times 10^{-3}}{5} = 0.1048$ seconds

↑ translation time

actuator speed $= 50 mm s^{-1}$

actuator stroke $= 100 mm$

assuming constant speed for the actuator

time $= 2$ seconds

↑ time for actuator to move

Total time $= 2(2) + 0.1048$

$= 4.1048$ seconds

Time for carrier to be transferred

Friction = 38.2N

~~weight~~ mass of carrier and rods = ~~90.94N~~ 9.273kg

Actuator stroke. 300mm

Output force of actuator: 1178N max at 6 bar

Limiting force of actuator to 300N

$$\Sigma Forces = ma$$

$$300 - 38\ 2 = 9.273a$$

$$a = 28.23\ ms^{-2}$$

Assuming constant acceleration

$$s = ut + \frac{1}{2}at^2 \qquad u = 0,\ v = 0$$

$$300 \times 10^{-3} = \frac{1}{2}(28\ 23)t^2$$

$$t = 0.02125 \text{ seconds}$$

Cycle time

Given that orientation and splitting can occur ~~simultaneously~~ simultaneously as lower carrier insertion

~~Cycle time:~~

time taken to fill buffer with 65 rods

$$\frac{65}{1.225} = 53.06 \text{ cycles} \rightarrow 54 \text{ cycles}$$

$$54 \times 2.45 = 132.3 \text{ seconds}$$

time taken to transfer lower carrier

$$= 0.02725$$

∴ Cycle time = 132.3 + 0.02725 + 4.1048

= 136.4 seconds

4.1048 → time taken for one row of rods to be inserted (final insertion does not occur in sync with other processes

Production Rate

$$\text{Production rate} = \frac{1}{\frac{136.4}{3600}}$$

PR = 26.4 modules per hour

Carrier Resupply

New carriers should be inserted after every module is complete

Given that the machine has a cycle time of 136.4 and there are 20 carriers in each feed.

$$20 \times 136.4$$
$$= 2728 \text{ seconds}$$

or

$$45.5 \text{ minutes}$$

∴ It is recommended to check the feed every 30 minutes, and resupply every 45.5 minutes to ensure constant flow of the machine.

Balancing Loss

$$B = \frac{N \times C - \sum_{n=1}^{N} t_n}{N \times C} \times 100\%$$

$C = 136.4$ seconds or 2.27 minutes

$N = 6$

$t_1 = 26.95$ seconds $\rightarrow$ 0.499 minutes
(Buffer and Splitting time)

$t_2 = 0$ (Splitting time assumed to be more significant)

$t_3 = 0.24$ seconds $\Rightarrow 4 \times 10^{-3}$ minutes
(lower corner insertion)

$t_4 = 20.5$ seconds $\rightarrow$ 0.3417 minutes
(All 65 rods to be inserted)

$t_5 = 0.021$ seconds $\Rightarrow 3.5 \times 10^{-4}$ minutes
(corner transfer)

$t_6 =$ negligible

$\therefore \quad B = 99.89\%$

Appendix F: Cost Calculations

Off The Shelf Component Costs				
Assembly	**Part Number**	**Part**	**Quantity**	**Cost £**
Rod Orientation	A2-1	Item 24 Extrusion 40x40 560mm	24	8.16
	A2-SA1-1	Mounting Plate	1	0.94
	A2-SA2-1	Monk Conveyers	1	1744.43
	A2-SA3-1	Monk Conveyers	1	375.54
	A2-SA4-1	Monk Conveyers	1	493.34
	A2-2	Item 24 Angle Bracket 40x40	19	8.79
	A2-3	Stepper Online PM stepper motor	3	3.23
	A2-SA1-5	Item 24 Conveyor	1	1288.67
	A2-4	Item 24 Adjustable Foot PA	27	5.74
	A2-5	Item 24 Extrusion 40x40 900mm	2	12.99
	A2-SA1-7	Optronis-camera	1	1000
	A2-SA2-2	Mounting Plate	1	0.94
	A2-SA2-3	Mounting Plate	1	0.94
	A2-6	Item 24 Cap 40x40	26	0.75
	A2-SA1-8	Washer M10	4	0.405
	A2-SA1-9	Item 24 Extrusion 40x40 210mm	1	3.03
	A2-SA2-6	WTB4S-3N2161 Sick Sensor	2	175.96
Rod Splitter	A3-1	Item 24 Extrusion 20 x 20 600mm	4	7.44
	A3-2	Item 24 Extrusion 20x20 320mm	6	3.968
	A3-3	Item 24 Conveyor	2	1653.61
	A3-4	Item 24 Extrusion 40x40 560	8	8.08
	A3-5	Item 24 Adjustable Foot PA	8	5.74
	A3-SA3-2	Item 24 Extrusion 40x40 360mm	3	5.19
	A3-6	Item 24 Angle Bracket 20	12	7.67
	A3-7	Stepper Online PM stepper motor	11	3.23
	A3-8	Item 24 Cap 20	20	0.52
	A3-9	Item 24 Cap 40	10	0.75
	A3-10	Item 24 Angle Bracket 40	15	8.79
	A3-SA3-3	Eriez Low Profile Magnetic Belt Coneyor	1	565.76
	A3-SA3-4	Sick Automation Light Grids	2	268.67
	A3-SA3-5	Light Gate Bracket	3	0.3
	A3-12	Sick Photoelectric Sensor	6	175.96
Rod Insertion	A4-SA2-3	SMC CDQM Compact Cylinder Guide Rod	1	105.84
	A4-SA2-3	Item Linear Unit KLE Line 5 40x40 LR	1	1000
	A4-SA2-5	Pushdown Actuator Mount	1	0.1755
	A4-SA2-6	Item Gearbox 40-3	1	50
	A4-SA2-7	Item Motor SE 40-035-3-90-AK-B	1	200
	A4-SA2-8	Item Drive Set KLE Line 5 40x40	1	50
	A4-SA2-9	Item Aluminium Extrusion Caps (40x40)	5	0.75
	AS 1110 - M5 x10	ISO metric hexagon precision bolts and screws	3	0
	AS 1110 - M4 x8	ISO metric hexagon precision bolts and screws	2	0
	A4-SA2-10	Sick Ultrasonic Proximity Sensor Mount	2	0
	A4-SA2-11	Sick Ultrasonic Proximity Sensor UC40-11211H	1	400
Bottom Carrier Input	A4-SA1-2	SMC CDQM Compact Cylinder Guide Rod	1	129.54
	A4-SA1-5	Item 40x40x62.5 Aluminium Extrusion	8	0.9020625
	A4-SA1-7	Item Adjustable Foot	10	5.74
	A4-SA1-8	Item Aluminium Extrusion Caps (20x20)	6	0.52
	A4-SA1-9	Item 45 Degree Bracket Cap	2	7.76
	A4-SA1-10	Festo DSBC-Standard cyliner: DSBC-50-300-PPVA-N3 (50mm B, 30mm Stroke)	1	185.06
	AS 1110 - M8 x12	ISO metric hexagon percision bolts and screws	2	0
	AS 1110 - M3 x 6	ISO metric hexagon percision bolts and screws	8	0
	AS 1110 - M4 x 8	ISO metric hexagon percision bolts and screws	4	0
	AS 1110 - M5 x 10	ISO metric hexagon percision bolts and screws	4	0
	A4-SA1-12	Sick Automation Light Grids MLG20N-0140E10501	1	268.67
	A4-SA1-13	Item 20x20x205.221 Aluminium Extrusion	4	2.5447404
	A4-SA1-14	Sick Ultrasonic Proximity Sensor Mount	2	0
	A4-SA1-4	Bottom Carrier Buffer	1	13.87576
	A4-SA1-15	Sick Ultrasonic Proximity Sensor UC40-11211H	2	400

Carrier Movement	A4-SA3-2	Festo DSBC-Standard cyliner: DSBC-50-300-PPVA-N3 (25mm Bore, 300mm Stroke)	1	61
	A4-SA3-8	Item Aluminium Extrusion Caps (40x40)	6	0.75
	A4-SA3-9	Sick Inductive Proximity Sensor IMS18-08BPSZC0SS03	1	138
	A4-SA3-10	Sick Ultrasonic Proximity Sensor Mount	1	0
	A4-SA3-11	Sick Ultrasonic Proximity Sensor UC40-11211H	1	200
	A4-SA3-6	Aluminium Frame	1	32.71093677
	A4-SA3-7	Full Legs	4	0.78991
	A4-SA3-12	WIKA F4801 Series Strain Gauge	1	62.82
Top Carrier	A4-SA4-M1-2	Matara Size 15 Unflanged Heavy Load Carriage	1	40.39
	A4-SA4-M1-3	Matara Size 15mm Heavy Load Counter Bored Linear Rail MSA Series	2	36.05
	A4-SA4-M1-4	SMC CDQM Compact Cylinder Guide Rod Body (63mm Bore, 40mm Stroke)	1	160
	A4-SA4-M1-5	SMC CDQM Compact Cylinder Guide Rod Body (63mm Bore, 40mm Stroke)	1	160
	A4-SA4-M1-7	SMC ISO Cylinder - CP96 (32mm Bore, 85mm Stroke)	1	76
	A4-SA4-M1-11	Sick Ultrasonic Proximity Sensor UC40-11211H Assembly	1	400
	AS 1110 - M8 x16	ISO metric hexagon precision bolts and screws	2	0
	AS 1110 - M5 x 16	ISO metric hexagon precision bolts and screws	4	0
	AS 1110 - M4 x 20	ISO metric hexagon precision bolts and screws	3	0
	A4-SA4-M2-1	Matara Size 15 Unflanged Heavy Load Carriage	1	40.39
	A4-SA4-M2-1	Matara Size 15mm Heavy Load Counter Bored Linear Rail MSA Series	2	36.05
	A4-SA4-M2-4	SMC ISO Cylinder - CP96 (32mm Bore, 140mm Stroke)	1	72
	AS 1110 - M6 x 12	ISO metric hexagon percision bolts and screws	8	0
	A4-SA4-M2-10	Sick Ultrasonic Proximity Sensor UC40-11211H Assembly	1	200
	AS 1110 - M6 x 12	ISO metric hexagon precision bolts and screws	8	0
	A4-SA4-M1-8	Red Support Structure	1	33.52361968
	A4-SA4-M2-8	Blue Support Structure	1	35.79546191
	AS 1110 - M3 x 30	ISO metric hexagon percision bolts and screws	4	0
Output Conveyor	A4-SA5-1	Habasit M1220 GripTop Conveyor Belt Assembly	1	1200
	A4-SA5-2	Item AC Conveyor Belt Motor	1	0
	A4-SA5-3	Item 16mm Shaft AC Motor Festening Set	1	0
	A4-SA5-4	Item Adjustable Foot	4	5.74
	A4-SA5-5	Item 40x40x28.65 Aluminium Extrusions	4	0.41350545
Input Conveyor	A1-2	Belt Conveyor Type GL-Straight Belt Conveyors	1	0
	A1-3	Item 24 Profile 12 60mm Extrusion 423mm	4	27.4
	A1-4	Item 24 Angle Bracket Set 6 60mmx60mm	4	11.09
	A1-5	Item 24 Profile 8 40mmx40mm Extrusion 400mm	2	5.77
	A1-6	Item 24 Angle Bracket 8 40mmx40mm	22	8.79
	A1-7	Item 24 Profile 8 40mmx40mm Extrusion 400mm	4	5.77
	A1-8	Item 24 Profile 40mmx40mm Extrusion 700mm	4	10.1
	A1-9	Item 24 Cap 40mm	12	0.75
	A1-10	Item 24 Cap 60mm	4	1.86
	A1-11	Item 24 Adjustable Foot PA 60mm	4	5.74

Custom Component Costs

Assembly	Part Number	Part	Quantity	Material	Cost per kg material	Mass	Machine Cost	Cost
Rod Orientation	A2-SA1-2	Rod Guding Channel	1	ABS Plastic	1.21	0.068	12	12.08228
	A2-SA1-3	Splitter	1	ABS Plastic	1.21	0.03	6	6.0363
	A2-SA1-4	Side Ramp	2	ABS Plastic	1.21	0.12	24	48.2904
	A2-SA1-6	Camera Bracket	1	Aluminium 6061-AHC	1.95	0.1944	10	10.37908
	A2-SA4-2	Rod Channel	1	Aluminium 6061-AHC	1.95	1.555	30	33.03225
	A2-SA2-4	Rod Blocking Fin	1	ABS Plastic	1.21	0.003	1	1.00363
	A2-SA2-5	2nd Rod Splitting Fin	1	ABS Plastic	1.21	0.01	2	2.0121
Rod Splitter	A3-SA1-1	Upper Splitter Channel	1	ABS Plastic	1.21	0.856	30	31.03576
	A3-SA2-1	Lower Splitter Channel	1	ABS Plastic	1.21	0.987	30	31.19427
	A3-SA3-1	Rod Buffer	1	ABS Plastic	1.21	1.17	200	201.4157
	A3-SA1-2	Fin Upper Splitter Front	1	ABS Plastic	1.21	0.0055	1	1.006655
	A3-SA1-3	Fin Upper Splitter Back	4	ABS Plastic	1.21	0.004	1	4.01936
	A3-SA2-2	Fin Lower Splitter Front	2	ABS Plastic	1.21	0.0056	1	2.013552
	A3-SA2-3	Fin Lower Splitter Back	4	ABS Plastic	1.21	0.0075	1	4.0363
	A3-11	Bracket	1	Aluminium 6061-AHC	1.95	0.0432	10	10.08424
	A3-13	Bracket for Photoelectic Sensor	6	Aluminium 6061-AHC	1.95	0.0324	10	60.37908
Rod Insertion	A4-SA2-1	Presser Assembly	1	ABS Plastic, Nickel, Stainless Steel				6.16675
	A4-SA2-2	Mounting Plate	2	Aluminium	1.95	0.216	10	20.8424
	A4-SA2-12	Aluminium Frame	1	Aluminium	1.95	22.63	20	64.1285
Bottom Carrier Input	A4-SA1-1	Lower Lift Plate	2	Stainless Steel	0.6	0.948	50	101.1376
	A4-SA1-3	Bottom Carrier Pusher	1	Stainless Steel	0.6	3.793	50	52.2758
	A4-SA1-11	Pushing Actuator Mount	1	Aluminium	1.95	0.972	30	31.8954
	A4-SA1-6	Push Up Actuator Mount	2	Aluminium	1.95	4.86	30	78.954
Carrier Movement	A4-SA3-1	Bottom Plate	1	Aluminium	1.95	2.948	30	35.7486
	A4-SA3-3	Carrier Pusher	1	Aluminium	1.95	0.693	30	31.35135
	A4-SA3-4	Actuator Mount 2	1	Aluminium	1.95	0.2592	30	30.50544
	A4-SA3-5	Actuator Mount 1	2	Aluminium	1.95	1.512	30	65.8968
Top Carrier Input	A4-SA4-M1-1	Top Clamp	1	Stainless Steel	0.6	4.55	50	52.73
	A4-SA4-M1-6	Mount	1	Aluminium	1.95	0.27	10	10.5265
	A4-SA4-M1-9	Red Actuator Bracket	2	Aluminium	1.95	0.506	10	21.9734
	A4-SA4-M1-10	Red Actuator Back Bracket	1	Aluminium	1.95	0.3	50	50.585
	A4-SA4-M2-3	Side L Plate	1	Stainless Steel	0.6	2.37	50	51.422
	A4-SA4-M2-5	Pusher Actuator Mount	1	Aluminium	1.95	6.48	10	22.636
	A4-SA4-M2-6	Red Actuator Bracket	2	Aluminium	1.95	0.0432	10	20.16848
	A4-SA4-M2-7	Red Actuator Back Bracket	1	Aluminium	1.95	0.0432	30	30.08424
	A4-SA4-M2-9	Pusher Actuator Mount Breackets	4	Aluminium	1.95	0.00405	10	40.03159
Output Conveyor	A4-SA5-6	Motor Foot	1	Various				5.74
Input Conveyor	A1-1	Rod Funnel	1	Aluminium 6061-AHC	1.95	2.106	30	34.1067
	A1-12	Ramp	1	ABS Plastic	1.21	0.072	14	14.08712

Total:	**20746.02681**		Item	Fasteners
			Sensors	Assemblies
			Actuators	Other